

MANDATE-primary 2026Q2 Empirical Evaluation: Engineering and Operational Provenance (Comprehensive Deposit Record)

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Abstract

This document is the comprehensive deposit record for the MANDATE-primary 2026Q2 empirical evaluation. It carries the full v1 substantive evidence (corpus authoring methodology, dataset construction, substantive findings catalog with verbatim model output samples, cross-system structural comparison, demo-era evidence) ALONGSIDE the project-internal engineering provenance: pilot smoke chronology, baseline calibration logs, apparatus patch history, per-phase cost log, full handoff chronology, realism audit machinery specification, and the v2 candidate patch verification trail.

Companion documents for academic submission:

- **Empirical Evidence Supplemental.pdf** — the focused academic supplement (substantive findings + cross-system comparison + structured Deviation Table + Reproducibility Checklist + Appendix X power analysis). Reviewers seeking only the v1 substantive evidence should read that document.
- **v2 Protocol Amendment.pdf** — the v2 methodology specification for re-evaluation under canonical MANDATE with full pre-registration coverage.

This document is preserved for replicators and the project’s deposit audit trail; the full engineering chronology supports independent re-execution but is not required reading for the manuscript’s empirical claims.

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1 Executive Overview

Document status — v1 evidence preserved (2026-06-23); v2 evaluation closed out 2026-07-08 under Deviations D-12/D-13 (see the Empirical Evidence Supplemental Deviation Table and the cost-ledger closeout addendum). This document records the v1 evaluation against the `mandate-eval-primary-2026q2-v1` freeze tag (commit 4f8af83) as completed. The five content-tripwire findings (Section 4) are evidenced at full coverage from on-disk RunRecord inspection and stand independent of Phase 8 grading.

On 2026-06-23 we discovered that the v1 Phase 8 grading rubric was authored against an array-of-dimensions `anchor.minimum/target` shape that does not match canonical MANDATE’s published schema (which specifies free-form objects). We verified this against the canonical `mlt-stack 1.0.0rc1` release. The v1 Phase 8 grading therefore measured *schema mismatch* rather than substantive coverage, and the cross-system numerical comparison in Section 5.4 is not interpretable as a head-to-head benchmark under that rubric. Deviation D-09 (Section 17) documents the v2 pivot: re-graded Cond-X (v1 records re-scored under the corrected v2 rubric) + Cond-A (canonical MANDATE on pre-extracted structured input) + Cond-B (canonical MANDATE with LLM-augmented Interpreter end-to-end). Deviations D-10 and D-11 (Section 17) record the subsequent PI directive to revoke the cost-driven Phase 8 sampling and judge substitutions and to reactivate the deferred O5 adversarial outcome: v2 grading is full pre-registration coverage (12,000 records, Opus ensemble, 20% IRR), and all 7 systems run the 350-perturbation suite at 10 runs each (24,500 perturbation records) graded under the same Opus ensemble. Section 5.4 is held open until v2 grading completes; until then the v1 numerical scores in that section are preserved as the rubric-mismatch-affected measurement that motivated the pivot, not as MANDATE performance claims.

The 9036 RunRecords, the corpus, the perturbation suite, the six baselines, the apparatus engineering chronology, and the demo-era qualitative findings are unchanged by the pivot and are reused under v2 per `handoffs/v2_salvage_audit.md` (42 KEEP / 8 REFRAME / 10 RE-GRADE / 38 RETIRE across 98 material assets).

The 2026Q2 evaluation produces 9036 RunRecords across seven systems on a 120-task source-conditioned main corpus and a 30-task out-of-domain hold-out. The principal substantive findings are:

1. **Decomposition single-COA prior reaffirmed at scale.** Across 1480 MANDATE-primary records (1180 successful main matrix + 300 hold-out), every record produces *exactly one* course of action. The canonical name (**Minimal manual assessment approach**) and two-node DAG shape (Manual Assessment → Results Documentation) are stable across all measured variation.
2. **Interpreter content-tripwire mode flip.** The Interpreter role’s `anchor.minimum.description` falls into one of two patterns: clean structured distillation or deterministic-prefix paragraph echo. Which mode is selected by chunk content fed to the upstream Procedure-role retriever, not by the user-facing request paragraph. Per-domain clean rates: 64.5% FIN, 56.0% INT, 72.5% SEC.
3. **Validator gap-acknowledgment instability.** The Validation role flags `anchor-vs-COA-count` gaps via heuristic phrases on 14–19% of records, inconsistently across otherwise-similar inputs.
4. **Binding structured refusal.** The Binding role emits `{"error": "..."}` when receiving contradictory inputs. Refusal rates: 2.0% FIN, 40.2% INT, 13.5% SEC, 7.0% out-of-domain hold-out. The INT rate inverts the demo-era prediction that financial content would dominate the refusal distribution.

5. **Intake content-tripwire.** On two tasks containing the natural-language phrase “Here’s the constraint:” followed by complex English, the **mandate-intake** fine-tune emits the English sentence as a constraint string, which fails the constraint grammar validator. 20 of 1200 main records (1.7%) affected; reproducible across all 10 seeds per task. Of 8 main-corpus tasks derived from the same source document (NIST SP 800-115), only the 2 carrying the trigger phrase fail.

The combined picture demonstrates that fine-tuned multi-role pipelines exhibit content-sensitive failure modes whose detection at scale requires a structured measurement apparatus and grounded ground truth. Five of six MANDATE roles show measurable content-tripwire behavior in v1-grounded evaluation. Section 5 quantifies the structural differences between MANDATE-primary’s mandate-as-code output and six baselines’ specification schemas; Section 6 documents the apparatus engineering required to measure these phenomena reliably.

Deposit state at v1 close (2026-06-23). The `outputs_freeze_v1_1` tag at commit `5f4de54` carries the 9036 RunRecords described in this document. The five-tag freeze chain (corpus, baseline, ground truth, perturbation, outputs) preserves the provenance integrity required for re-evaluation. Phase 8 three-judge grading completed on the 700-record D-08 stratified sample with HALT under PROTOCOL_LOCK §8 (min pairwise $\kappa = 0.296 < 0.40$). Subsequent analysis identified the rubric-shape mismatch motivating Deviation D-09 (v2 pivot, Section 17). The v2 deposit will carry the **mandate-eval-primary-2026q2-v2** tag with three measured conditions (Cond-X / Cond-A / Cond-B), the corrected rubric, full pre-registration grading coverage under Opus (D-10), the reactivated O5 perturbation outcome (D-11), and the quantified schema-mismatch penalty; v1 evidence in this document remains the v1 record of measurement.

2 Methodology

2.1 Why Source-Conditioned Corpus Authoring

The 2026-06-03 reconciliation moved PROMPTS Section 1 from synthetic free-form task generation to source-first source-conditioned generation. Each candidate operational request is derived from a real public-source chunk sampled deterministically from a domain-specific Jaccard-indexed corpus. The authoritative source set carries 18 documents across the four corpus domains. Every candidate carries a **derived_from** reference to the chunk it was generated against, which lets the evaluation verify substantive content against authoritative documents and prevent fabricated thresholds from leaking into the test set.

The four corpus domains and source-document inventory:

- *security_operations_reporting* (7 sources): NIST SP 800-53 Rev. 5, NIST SP 800-61 Rev. 2, NIST SP 800-115, NIST SP 800-137, NIST SP 800-92, MITRE D3FEND public knowledge graph, CISA Cybersecurity Advisories index.
- *financial_reporting* (8 sources): Sarbanes-Oxley Act, NIST SP 800-37 Rev. 2 Risk Management Framework, OMB Circular A-123, COSO Internal Control Integrated Framework, PCAOB AS 2110, AS 2201, AS 2410, SEC Form 10-K general instructions.
- *intelligence_collection_tasking* (3 sources): ODNI Annual Threat Assessments 2024 and 2025, intelligence.gov IC overview.

- *software_engineering_specification* (hold-out, 7 sources fetched of 9 curated): NIST SP 800-160 Vol. 1, NIST SP 800-218, NIST SP 800-64 Rev. 2, Agile Manifesto, Twelve-Factor App, NASA NPR 7150.2D, UK GDS Service Standard.

Source documents are extracted from their original binary forms (PDF, HTML, DOCX, PPTX) through the apparatus’s production extractors and indexed as Jaccard chunks with chunk size 1200 tokens and overlap 200 tokens. Candidate generation uses Claude Opus 4.6 at temperature 0; chunk sampling uses deterministic seed 20260603. The full source list and fetch report (recording the HTTP outcome of every URL attempt) is in `code/rag/embeddings/build_report.json`.

2.2 Why Six Baselines

Pre-registered baselines span the orchestration patterns a team would plausibly build today against the same evaluation tasks:

- **B1: single-prompt Claude Sonnet 4.6**, no RAG, no role decomposition. The simplest possible baseline.
- **B2: single-prompt GPT-4o**. Same structure as B1, different model family.
- **B3: ReAct loop on Claude**. Tool-use agentic loop without explicit JSON-mode enforcement (the protocol does not permit enforcement; it would change the system under measurement).
- **B4: AutoGen PlannerReviewer shape**. Multi-agent: a planner proposes, a reviewer critiques, planner revises.
- **B5: CrewAI SequentialCrew shape**. Multi-agent: a sequence of role-specialized agents.
- **B6: LangGraph GraphRevision shape**. Multi-agent: explicit graph orchestration with revision loops.

B4-B6 are framework-pattern shells implemented in apparatus code (the direct LLM-client calls follow the framework’s orchestration shape deterministically) rather than literal AutoGen/CrewAI/LangGraph installations. The intent: the framework is the variable, the model is held constant (Claude Sonnet 4.6 across B1, B3, B4-B6). This isolates the orchestration pattern’s measurable effect rather than confounding model and orchestration choice.

2.3 Why Three Judges

Phase 8 grading uses an ensemble of three LLM judges from three vendor families: GPT-4o (`gpt-4o-2024-11-20`), Claude Opus 4.6 (`claude-opus-4-6`), and Gemini 2.5 Pro (`gemini-2.5-pro`). Three judges from three different families maximize independent perspective and support an inter-judge reliability check. Cohen’s kappa below 0.40 on any judge pair triggers protocol halt (PROTOCOL_LOCK §8). A 20% sample (deterministically seeded with 20260616) is double-graded to support the IRR computation; the apparatus patch enabling the double-grade pass landed mid-execution (see Section 6.3).

The Judge token budgets differ by family. GPT-4o and Claude Opus 4.6 use `max_tokens=2048`. Gemini 2.5 Pro uses `max_tokens=8192` because the model’s thinking-mode reasoning tokens count against the output budget before the visible structured-score JSON; the 2048 budget produces empty visible output (mid-grading diagnosis). The 8192 budget costs roughly \$0.04 per call vs the 2048-budget GPT-4o calls at \$0.02 each.

2.4 The SME-Skip Deviation

The original pre-registration specified an SME (subject-matter expert) realism audit between corpus selection and ground-truth freeze, plus an SME accept/edit/reject step for the PROMPTS Section 2 anchor scaffolds. The principal investigator authorized the deviation recorded in `code/00_preregistration/DEVIATIONS.md` (2026-06-04) to skip SME involvement for this evaluation. Under the deviation, the PROMPTS Section 2 anchor scaffolds produced by Claude Opus 4.6 serve as the ground truth verbatim. This means:

- Phase 8 grading measures system-vs-Claude-Opus-scaffold alignment, not system-vs-SME-truth alignment.
- Cohen’s kappa under the SME-skip condition measures inter-judge agreement on a Claude-Opus-generated answer key, not inter-judge recovery of SME judgment. The halt threshold (0.40) remains binding but the semantic interpretation shifts.
- Realism is an acknowledged open caveat. Reviewers asking “is the corpus realistic?” receive the honest answer: the corpus is the source-conditioned output of Claude Opus 4.6 against curated public-source chunks; realism was not SME-adjudicated in this exercise.

A future v2 evaluation with SMEs is forward-compatible: the existing scaffolds become SME-review-candidate anchors, the existing Phase 6 RunRecords get re-graded against the SME-revised ground truth, and a `gt_freeze_v2` tag layers cleanly atop the v1 baseline.

3 Dataset Construction and Statistics

3.1 Phase 6 Matrix and Freeze-Tag Chain

Phase 6 produces 9036 RunRecords distributed as:

System	Set	Records
MANDATE-primary	main	1200
Baseline 1 (single-prompt Claude)	main	1206
Baseline 2 (single-prompt GPT-4o)	main	1206
Baseline 3 (ReAct Claude)	main	1206
Baseline 4 (AutoGen PlannerReviewer)	main	1206
Baseline 5 (CrewAI SequentialCrew)	main	1206
Baseline 6 (LangGraph GraphRevision)	main	1206
MANDATE-primary	hold-out	300
Baseline 1	hold-out	300
Total		9036

The baselines wrote 1206 instead of the targeted 1200 each due to 6 over-budget records per baseline retained from earlier reruns; these are preserved rather than truncated. The full inventory is at `standalone data results/dataset_inventory/record_counts.md`.

The five-tag freeze chain:

Tag	Commit	Date	Covers
<code>corpus_freeze_v1</code>	05fc0aa	2026-06-04	Task JSONLs
<code>baseline_freeze_v1</code>	05fc0aa	2026-06-04	B1-B6 calibration
<code>gt_freeze_v1</code>	068fe94	2026-06-05	Anchor scaffolds
<code>perturbation_freeze_v1</code>	068fe94	2026-06-05	350-perturbation suite
<code>outputs_freeze_v1_1</code>	5f4de54	2026-06-15	Phase 6 RunRecords corrected

The original `outputs_freeze_v1` is preserved in the git log as the historical record of the 300 contaminated MP hold-out records quarantined and regenerated after the hold-out contamination correction (see Section 6.5).

3.2 Per-Domain Phase 6 MANDATE-Primary Main Matrix Observations

Across 1200 MANDATE-primary main-matrix records, the substantive per-role counts by domain:

Domain	Records	OK	Intake-fail	Binding refusal	Interp. clean	Val. gap-flag
financial	400	400	0	8 (2.0%)	258 (64.5%)	77 (19.2%)
intel-collect	400	400	0	161 (40.2%)	224 (56.0%)	56 (14.0%)
sec-ops	400	380	20	54 (13.5%)	290 (72.5%)	77 (19.2%)
Overall	1200	1180	20	223 (18.6%)	772 (64.3%)	210 (17.5%)

3.3 MANDATE-Primary Wall-Clock Distributions per Domain

All values in milliseconds. “Mean (intake fail)” applies only to SEC where the 20 fast Intake failures pull the mean down.

Domain	n	min	p25	median	p75	max	mean (ok only)	mean (intake fail)
FIN	400	91,176	101,457	105,498	110,280	229,652	109,884	—
INT	400	90,579	98,861	103,171	107,614	126,558	103,262	—
SEC	400	7,886	103,889	107,426	111,733	130,263	108,844	8,404

Median MANDATE-primary main wall-clock is approximately 105 seconds per run. The SEC minimum of 7,886 ms is the fast-Intake-failure path (approximately 8 seconds for the Intake LLM call plus the constraint grammar validation reject); the SEC mean (intake fail) confirms this. The FIN maximum of 229,652 ms is a single tail-end run; the p75 of 110,280 ms indicates 75% of runs complete within 110 seconds.

3.4 Hold-Out Generalization

The 30-task hold-out is in the fourth corpus domain (*software_engineering_specification*), deliberately outside the distribution of all six MANDATE role fine-tunes. The 300 MANDATE-primary hold-out records were regenerated after the contamination correction when the original 300 were contaminated by an Ollama outage (Section 6.5):

- **ok rate:** 300/300 (100%).
- **COA count distribution:** all 300 records emit exactly one COA. The Decomposition single-COA prior generalizes to the out-of-domain setting.

- **Binding refusal rate:** 21/300 (7.0%). Between the in-distribution FIN (2%) and SEC (13.5%) rates, well below INT (40%). The Binding fine-tune generalizes more robustly to SE-spec content than to in-distribution intelligence-collection content.

4 Substantive Findings Catalog

This is the core of the document. Each finding includes the substantive claim, the Phase 6 evidence, verbatim model output samples, and the training-data implications for the upstream MANDATE team.

4.1 Finding 1: Decomposition Single-COA Prior on Unstructured Input

Update (2026-06-23)

This finding is preserved as the v1 observation. Subsequent empirical verification against canonical MLT-Governance-Stack v1.0.0rc1 MANDATE (handoffs/MLT_realness_audit_opus.md) showed that the single-COA two-node-DAG output (**Manual Assessment** → **Results Documentation**) is the canonical MANDATE fallback path when the **MissionInput** carries no **available_tools**, no **constraints**, and an empty **scope**. Under structured input (e.g., the shipped **tests/examples/normal_mission.json** with three tools and four constraints) the same canonical MANDATE produces three COAs (Conservative / Moderate / Aggressive) with multi-node DAGs. The v1 evaluation fed natural-language task text into MANDATE without first extracting **available_tools** or **constraints**, so the v1 records measure “MANDATE’s behavior given unstructured input,” not “MANDATE’s decomposition capability per se.” Cond-A under v2 pre-extracts structured **MissionInput** before calling MANDATE and will measure decomposition capability directly.

Claim

The Decomposition role emits exactly one course of action when invoked with unstructured natural-language input. Across 1480 successful MANDATE-primary records under the v1 setup, every record produces `n_coas == 1`.

Evidence

COA-1 name distribution across all ok records:

COA-1 name	Count
Minimal manual assessment approach	1480

The same canonical name appears in every record. The two-node DAG shape (**Manual Assessment** → **Results Documentation**) is also stable across all 1480 records.

DAG node description specificity varies by domain

The COA count is stable but DAG node descriptions vary qualitatively between domains. A sample of distinct descriptions (the full list is at **standalone data results/finding_1_decomposition/dag_descriptions_by_domain.md**):

Security operations (specific):

Manual assessment of MITRE ATT&CK technique hardening, SOC
detection integration, IT/OT segmentation validation

Financial reporting (specific):

Manual assessment of COSO ERM framework alignment, NIST SP
800-37 risk management, Federal Reserve supervisory expectations

Intelligence collection (often generic):

Manual assessment of target scope

Hold-out (generic):

Manual assessment of target scope
Document assessment findings

The Documentation node text (“Document assessment findings”) is identical across all 1480 records.

Interpretation revised post-canon-verification

The v1 reading of this evidence (“the Decomposition fine-tune has learned a strong single-COA prior”) was partially correct but conflated two phenomena. The two phenomena, separated post hoc:

- (a) *Canonical fallback path under unstructured input.* The canonical Decomposition role default-templates a single `Minimal manual assessment approach` COA when `available_tools` is empty, `constraints` is empty, and no `DomainProfile` is configured. This is implemented behavior in `mlt-stack 1.0.0rc1` and reproduces under canonical MANDATE on the v1 corpus (empirical verification, 2026-06-23). The v1 evaluation fed natural language without first extracting these structured fields, so 1480/1480 records hit the fallback path.
- (b) *Open question: does the fine-tune fan out under structured input?* Whether the `mandate-decomp 32B Qwen3` fine-tune generates multiple COAs when given structured input with explicit multi-option requirements (the demo-era CrowdStrike and SVB CIO/CFO “three distinct strategic options” prompts both produced `n_coas=1` even with structured-looking input) remains open and is the subject of Cond-A and Cond-B measurement in v2; Deviation D-09 (Section 17) specifies the v2 test design.

The upstream-team note’s training-data recommendation (audit Decomposition seed corpus for multi-option anchors) is retained as forward-looking guidance pending Cond-A results.

Reproduction

```
import json, glob, collections
ns = collections.Counter()
for f in glob.glob("artifacts/07_system_outputs/mandate_primary/*.json"):
    r = json.load(open(f))
    art = (r.get("output") or {}).get("artifact") or {}
    coas = art.get("courses_of_action") or []
    ns[len(coas)] += 1
print(ns) # expected: Counter({1: 1180, 0: 20})
```

4.2 Finding 2: Interpreter Content-Tripwire Mode Flip

Claim

The Interpreter role’s `anchor.minimum.description` falls into one of two patterns: clean structured distillation, or deterministic-prefix paragraph echo. The mode is selected by chunk content fed to the upstream Procedure-role retriever, not by the user-facing request paragraph.

Evidence: Per-Domain Mode Rates

Domain	Clean distillation	Deterministic-prefix	Clean rate
financial_reporting	258	142	64.5%
intelligence_collection_tasking	224	176	56.0%
security_operations_reporting	290	90	72.5%
Overall	772	408	65.4%

Verbatim sample: clean distillation (FIN)

Three distinct strategic options for endpoint security architecture with (a) operational and capex implications, (b) residual risk profile after one year, (c) vendor-relationship and contractual changes, and (d) implementation timeline.

Verbatim sample: deterministic-prefix paragraph echo (FIN)

Minimally satisfy: Team, this is the CFO. The audit committee has asked for a strategy memo following the Silicon Valley Bank collapse of March 2023. We need three options for ERM overhaul that address treasury management, deposit concentration risk, and AOCI exclusion. Strategy memo deliverable by board meeting in 30 days.

Mode flip correlates with source document

A per-source-document analysis (from `standalone data results/finding_2_interpreter/mode_by_source_document.md`) shows the clean rate varies substantially by source. Selected source documents:

Source Document	Clean	Det-prefix	Clean rate
NIST SP 800-53 Rev. 5	96	24	80.0%
NIST SP 800-37 Rev. 2 RMF	82	38	68.3%
COSO Internal Control Integrated Framework	47	13	78.3%
NIST SP 800-115	64	16	80.0%
ODNI ATA 2024	78	32	70.9%
SEC Form 10-K General Instructions	22	28	44.0%
ODNI ATA 2025	56	64	46.7%

Sources where the chunk-shape evidently favors clean distillation include NIST SP 800-53 and NIST SP 800-115 (both at 80%). Sources where the chunk-shape evidently triggers deterministic-prefix more often include SEC Form 10-K (44%) and ODNI ATA 2025 (47%). The same fine-tuned model produces both modes across the same evaluation; chunk content selects the mode.

Training-data implication

The Interpreter fine-tune has both output modes reachable from the same model state, with chunk-content patterns selecting between them. Adding chunk-shape-aware curriculum to the training data (or per-example labels indicating the intended distillation style) would let the fine-tune learn which shape to emit per chunk pattern. The upstream-team note’s strongest interpretation: the financial-domain fine-tune’s anchor distillation in the original demo work was content-tripwire-driven, not domain-bound; the same applies cross-domain at Phase 6 scale.

4.3 Finding 3: Validator Gap-Acknowledgment Instability

Claim

The Validation role flags anchor-vs-COA-count gaps via heuristic phrases (**insufficien**, **potential**, **may not**) on 14–19% of records, inconsistently across otherwise-similar inputs.

Evidence

Domain	Gap-flagged	Rate
financial_reporting	77 / 400	19.2%
intelligence_collection_tasking	56 / 400	14.0%
security_operations_reporting	77 / 400	19.2%
Overall	210 / 1180	17.8%

Verbatim sample: flagged rationale

COA-1 is selected as the primary course of action because it presents the lowest risk with a score of 'LOW' and aligns with the current resource constraints. However, its low confidence in execution and potential insufficiency in delivering three distinct strategic options necessitate close monitoring and possible adjustments.

Verbatim sample: unflagged rationale (similar input)

COA-1 chosen for resource efficiency despite low confidence; aligns with minimal assessment requirements while acknowledging execution risks.

Both rationales describe the same single-COA-vs-three-options gap; only the first one explicitly flags it. The full samples set is at `standalone data results/finding_3_validator/`.

Training-data implication

The Validation role’s gap-detection appears surface-pattern-driven rather than constraint-count-driven. The fine-tune recognizes specific verbal patterns (**may not**, **insufficien**, **potential**) but does not reliably compute the anchor’s **minimum** cardinality against the actual COA count. Training data that explicitly labels the cardinality-mismatch signal would improve detection consistency.

4.4 Finding 4: Binding Structured Refusal

Claim

When the Binding role receives an anchor that demands more courses of action than Decomposition emitted, the **mandate-binding** fine-tune emits a valid JSON structured refusal: a single field **error** containing a natural-language explanation. The v1 apparatus parses this as a parse failure (the schema requires a **decision_summary** field which the refusal omits) and falls back to the deterministic Binding path. The v2 candidate patch routes the refusal to **output.gap_reports** instead.

Per-Domain Refusal Rates

Domain	Records	Binding refusal	Rate
financial_reporting	400	8	2.0%
intelligence_collection_tasking	400	161	40.2%
security_operations_reporting	400	54	13.5%
software_engineering_specification (hold-out)	300	21	7.0%
All Phase 6 MP	1500	244	16.3%

The INT rate at 40% is roughly $20\times$ the FIN rate (2%). The hold-out rate at 7% is between the in-distribution FIN and SEC rates, indicating that the Binding fine-tune’s refusal behavior generalizes to out-of-domain content but with content-dependent rate variation that does not simply correlate with domain familiarity.

Verbatim refusal samples from the demo era

Three consecutive direct-Ollama calls against the SVB demo task all returned the same structural refusal pattern. Verbatim, attempt 1:

```
{"error": "The provided JSON data does not contain the necessary information to make a recommendation. The 'coas' array is empty, and there is only one COA available (COA-1), which may not meet the requirement for three distinct strategic options as mentioned in the mission intent."}
```

Verbatim, attempt 2:

```
{"error": "The provided JSON data does not contain the necessary information to make a recommendation. There is only one COA (COA-1) available, but no alternative options to consider for a fallback sequence. Additionally, there is no risk tolerance specified in the anchor to guide the recommendation."}
```

Verbatim, attempt 3:

```
{"error": "The provided content does not meet the requirements outlined in the mission intent. Specifically, there is only one COA (COA-1) presented, while the mission explicitly requests three distinct strategic options. Additionally, the provided COA lacks the necessary details regarding operational/capex implications, residual risk profiles, contractual/personnel changes, and implementation timelines as required."}
```

The three error messages share a structural pattern: the model identifies the cardinality mismatch (1 COA vs anchor’s demand for 3) and declines to fabricate the `decision_summary` field over the contradiction.

Training-data implication

The Binding fine-tune is correctly refusing to fabricate a `decision_summary` over contradictory upstream inputs. This is honest model behavior. The v1 apparatus discards the `error` text after 3 retries; the v2 candidate patch (`feature/binding-refusal-as-gap-sideload`) routes the refusal to `output.gap_reports`. The upstream-team note’s strongest interpretation: the Binding training data does not need additional financial-domain examples to “fix” the refusal; the apparatus needs to listen to it. The Phase 6 hold-out generalization at 7% supports the training-data-is-sufficient interpretation.

4.5 Finding 5: Intake Content-Tripwire (“Here’s the constraint:”)

Claim

The Intake role’s `validate_constraint()` requires the MANDATE constraint grammar (`field operator value` with operators `==`, `!=`, `<`, `<=`, `>`, `>=`, `IN`, `CONTAINS`). On two main-corpus tasks (TASK-MAIN-SEC-038 and TASK-MAIN-SEC-040, both `stretch_case` category, both derived from NIST SP 800-115), the request text contains the natural-language phrase “Here’s the constraint:” followed by complex English. The `mandate-intake` fine-tune treats the phrase as a directive: it emits the subsequent English sentence verbatim as a constraint string, which fails the grammar validator. The pipeline halts at role 1.

All 20 Intake error texts

All 20 records show essentially identical error text, demonstrating that the failure is deterministic across all 10 seeds for each affected task. A representative sample (the full enumeration is at `standalone data results/finding_5_intake/all_20_intake_errors.md`):

```
[r01] Intake: Invalid constraint syntax: [0] the full assessment engagement, including all six data centers, needs to wrap up and have final reports delivered within two calendar weeks

[r05] Intake: Invalid constraint syntax: [0] the full assessment engagement, including all six data centers, needs to wrap up and have final reports delivered within two calendar weeks

[r10] Intake: Invalid constraint syntax: [0] the full assessment engagement, including all six data centers, needs
```

to wrap up and have final reports delivered within two calendar weeks

All 10 of SEC-038’s runs produce byte-identical error text; the phenomenon is fully deterministic given the seed family (seed_base 20260605, run_number 1..10).

Critical control: the trigger is the phrase, not the source document

A potential confounder: both affected tasks derive from NIST SP 800-115. Could the source document be the trigger rather than the phrase?

Eight main-corpus tasks derive from NIST SP 800-115. The Intake failure rates per task:

Task ID	Intake failure rate
TASK-MAIN-SEC-005	0/10
TASK-MAIN-SEC-008	0/10
TASK-MAIN-SEC-013	0/10
TASK-MAIN-SEC-022	0/10
TASK-MAIN-SEC-035	0/10
TASK-MAIN-SEC-038	10/10
TASK-MAIN-SEC-039	0/10
TASK-MAIN-SEC-040	10/10

Six of eight tasks derived from NIST SP 800-115 produce zero Intake failures. Only the two carrying the “Here’s the constraint:” phrase fail. The trigger is the phrase, not the source document. This sharpens the training-data implication: the Intake fine-tune needs examples that distinguish “the constraint” as part of the user’s prose from “the constraint” as a directive to emit one in MANDATE grammar.

Training-data implication

Audit the Intake fine-tune’s training set for examples where the user request text contains the literal word “constraint” or “constraints” embedded in natural language. The fine-tune appears to lack examples that distinguish these usages. Three apparatus-side or training-data mitigations are documented in DEVIATIONS.md (Section 7): (1) make `Intake.execute()` log invalid constraints as a metric without failing the role, (2) patch the Intake prompt to be explicit that “the constraint” in user text is not a directive, (3) replace SEC-038 and SEC-040 from the 142-candidate residue of the 262-candidate main corpus pool. The v1 study runs with the failure mode observed.

4.6 Finding 6: Schema-Mismatch Effect in v1 Phase 8 Grading (Quantification Pending v2)

Claim

The v1 Phase 8 grading rubric was authored against a non-canonical `anchor.minimum/target` schema (arrays of `{dimension, threshold, rationale}` dicts). Canonical MANDATE (`mlt-stack 1.0.0rc1`, verified REAL by three independent auditors and a direct end-to-end pipeline run, 2026-06-23) emits `anchor.minimum` and `anchor.target` as free-form *objects* with one-or-more key-value pairs. Three SOTA judges (GPT-4o, Claude Sonnet 4.6, Gemini 2.5 Pro) graded structurally valid canonical MANDATE outputs as 0.0 coverage because the system’s

single-key minimum object did not match the array-of-dimensions shape the rubric expected. The v1 cross-system numerical comparison in Section 5.4 is therefore a measurement of schema mismatch, not substantive MANDATE performance.

Evidence

- **Canonical schema.** `mandate-as-code.schema.json` in MLT v1.0.0rc1 specifies `anchor.minimum` as `{type: "object", minProperties: 1, additionalProperties: oneOf[number, string, boolean, object, array]}`. No requirement for `dimension/threshold/rationale` keys.
- **Empirical canon-vs-v1 shape match.** Five test tasks (TASK-MAIN-INT-034 CW threat matrix, TASK-MAIN-FIN-001, FIN-018, INT-003, SEC-014) run through canonical MLT MANDATE produce identical shape to the v1 MANDATE-primary records on the same tasks: `anchor.minimum = {description: <string>}`, one COA, two-node DAG. The v1 records are valid canonical outputs.
- **Per-system ensemble scores under v1 rubric.** `mandate_primary: minimum_coverage` mean 0.179, `constraint_coverage` mean 0.010, gap TP 0/100. Baselines 1/3/4/5/6: `minimum_coverage` 0.85–0.96, gap TP 100/100. The Sonnet judge delta on MANDATE-primary was -0.301 vs the other two judges but $+0.04$ to $+0.08$ on every baseline, consistent with shape-bias against the canonical MANDATE schema rather than generation-family bias.
- **Kappa halt was on mission_intent_match.** v1 main-pass min pairwise $\kappa = 0.296 < 0.40$ threshold (PROTOCOL_LOCK §8), driven by Sonnet-Gemini disagreement on `mission_intent_match`. Per-outcome Krippendorff's α shows the failure is on subjective outcomes (`mission_intent_match` 0.299, `trace_completeness` 0.194, `fabrication_count` 0.216), not objective ones (`gap_classification` 0.701, `minimum_coverage` 0.612).

v1 reading vs revised reading

The v1 reading of the Phase 8 results was “MANDATE-primary underperforms baselines on coverage outcomes.” The revised reading, informed by the schema-mismatch discovery, is “the v1 rubric scored schema shape match, not semantic coverage; under a shape-agnostic rubric the cross-system comparison is unmeasured.” The five content-tripwire findings in Sections 4.1–4.5 are unaffected: they are evidenced from on-disk RunRecord inspection (verbatim quotes, structural counts, cross-extraction-path comparisons) and do not depend on Phase 8 numerical grading.

Quantification under v2

Deviation D-09 (Section 17) specifies the v2 three-condition evaluation:

- **Cond-X re-graded.** The 100 v1 MANDATE-primary records in the D-08 sample are re-graded under the corrected v2 rubric (semantic coverage of ground-truth dimensions against canonical free-form-object minimum). The per-record delta $v2 - v1$ quantifies the schema-mismatch penalty.
- **Cond-A.** 1500 records produced by canonical MLT MANDATE v1.0.0rc1 against pre-extracted structured `MissionInput`. Tests MANDATE planning capability isolated from natural-language extraction.

- **Cond-B.** 1500 records produced by canonical MLT MANDATE with LLM-augmented Interpreter end-to-end on raw natural-language input. Tests MANDATE as integrated system.
- Baselines B1–B6 re-graded under the v2 rubric for apples-to-apples cross-system comparison.

Headline result placeholder: *schema-mismatch penalty on minimum_coverage = XX.XX points (Cond-X v1 vs Cond-X v2)*; populated when v2 grading completes.

Methodological contribution independent of MANDATE performance

The quantified schema-mismatch penalty is publishable as a methodology finding regardless of how Cond-A and Cond-B compare to baselines: it shows that LLM-as-judge ensembles will systematically penalize structurally novel outputs against ground truth generated by a different schema family, and the penalty is large enough to produce a `PROTOCOL_LOCK` κ halt on subjective outcomes even when objective outcomes (gap classification, schema validity) show acceptable agreement.

4.7 Cross-Role Synthesis: Five of Six Roles Show Content-Tripwires

The Procedure role is the only role uncharacterized for content-tripwire behavior in this dataset (the role consumes the upstream RAG retriever’s chunks and produces process descriptions; pure pass-through with retrieval-shaped output). The combined picture:

Role	Content-tripwire characterized	Evidence at scale
Intake	yes (this work)	20 / 1200 main records
Interpreter	yes (demo + this)	408 / 1180 det-prefix mode
Decomposition	yes (single-COA prior)	1480 / 1480 records
Procedure	no (uncharacterized)	—
Binding	yes (this work and demo)	244 / 1500 refusals
Validation	yes (instability)	210 / 1180 flagged

The upstream-team training-data audit recommendation strengthens: chunk-shape diversity, surface-pattern diversity, and constraint-cardinality diversity should be sampled in the training data, not just request-paragraph diversity. The v1 evaluation reported here measures the phenomenon; the v2 candidate apparatus patches and training-data mitigations are documented for follow-up.

Substantive Synthesis from the Demo Memo

The demo-era findings memo (`demo/MANDATE_DEMO_FINDINGS.md`, 2026-06-03 author: Lead Analyst) recorded the four original findings before Phase 6 scaled them. The memo’s principal conclusion bears quoting:

“MANDATE-primary’s apparatus is verified end-to-end on three real, source-grounded scenarios across two domains. Every fine-tuned role fires without silent fallback, RAG retrieval is wired into the Procedure role, the schema-compliant mandate-as-code artifact emits cleanly, and the trace chain is intact.”

“The Decomposition role currently produces one COA per request, cross-domain. Three scenarios, two domains (security operations on Volt Typhoon and CrowdStrike, financial reporting on SVB), every Ollama run produces a single COA with the same name and

the same two-node DAG, even when the anchor’s minimum explicitly demands three. This is a structural tendency in the fine-tune, not a scenario-dependent artifact, and not an apparatus defect.”

The memo’s framing on training-data implications, also worth quoting:

“Add training examples whose anchors carry explicit multi-option requirements (a CISO request asking for three strategy variants is a common operational shape) and check whether the Decomposition output fans to N COAs when the anchor requires N. Consider whether the Decomposition prompt template explicitly receives the anchor’s minimum cardinality requirement. If the prompt summarizes the anchor before passing it forward, that summary may be lossy with respect to the ‘N distinct options’ signal.”

The Phase 6 evidence at 1480 records confirms the demo memo’s hypothesis: the single-COA prior is robust to all measured input variation. The training-data audit recommendation is the actionable upstream-team work this evaluation supports.

5 Cross-System Structural Comparison

5.1 Anchor Density Across Seven Systems

MANDATE-primary uses the mandate-as-code schema where `anchor.minimum` is a single structured object with a `description` field. The six baselines use the `baseline_specification` schema where `minimum` is an array of `{dimension, threshold, rationale}` objects. The following table reports the median number of entries per record across `ok=True` records:

System	<i>n</i>	Schema-valid	Min med	Min mean	Tgt med	Con med	Gap med	Gap max
<code>mandate_primary</code>	1180	n/a	1.0	1.00	1.0	0.0	0.0	0
<code>baseline_1</code>	1206	1201/1206	8.0	8.23	4.0	6.0	9.0	13
<code>baseline_2</code>	1206	1205/1206	3.0	3.16	1.0	2.0	2.0	5
<code>baseline_3</code>	1205	0/1205	8.0	8.67	6.0	6.0	10.0	25
<code>baseline_4</code>	1206	1195/1206	8.0	8.22	4.0	6.0	10.0	15
<code>baseline_5</code>	1206	1196/1206	7.0	7.28	5.0	7.0	10.0	15
<code>baseline_6</code>	1206	1197/1206	8.0	8.24	4.0	6.0	9.0	14

Three observations:

(i) **Structural difference between MANDATE-primary and baselines is large.** Baselines emit 3–8 structured minimum entries per record; MANDATE emits 1 (a single `anchor.minimum` object with one `description` field). On the demo-era CrowdStrike comparison, single-prompt Claude (B1) produced 9 structured minimum entries with `dimension` and `threshold` fields versus MANDATE’s 1 free-text paragraph; the Phase 6 1206-record measurement confirms the demo finding at scale.

(ii) **B3 schema-validity is 0/1205.** ReAct on Claude produces 8.67 minimum entries on average but 0 schema-valid outputs across 1205 records. The ReAct loop converges on structurally-flat JSON (arrays of strings, keyed objects in place of arrays of objects) that does not match the locked `baseline_specification` schema. Without explicit JSON-mode enforcement (the protocol does not permit it; would change the system under measurement), this is B3’s characterized behavior at scale.

(iii) **Suspected-gaps richness varies dramatically.** Baseline 1 produces a median of 9 suspected gaps per record. Baseline 2 (single-prompt GPT-4o) produces only 2. MANDATE-primary produces 0 inside the anchor (MANDATE’s mandate-as-code schema carries gaps in `output.gap_reports`, not in the anchor). The baseline-vs-MANDATE comparison is therefore not apples-to-apples on gap counts; Section 5.3 explains the mandate-as-code output structure.

5.2 Per-Baseline Schema-Validity per Domain

The B3 0/1205 schema-validity result is stable across domains:

Baseline	FIN valid	INT valid	SEC valid
baseline_1 (single-prompt Claude)	399/400	400/400	402/406
baseline_2 (single-prompt GPT-4o)	400/400	400/400	405/406
baseline_3 (ReAct Claude)	0/400	0/400	0/405
baseline_4 (AutoGen)	397/400	397/400	401/406
baseline_5 (CrewAI)	397/400	396/400	403/406
baseline_6 (LangGraph)	398/400	398/400	401/406

All non-B3 baselines hit at least 99% schema-valid in every domain. B3’s 0% is uniform across all three domains; the ReAct framework’s structural failure does not vary by content domain.

5.3 MANDATE Mandate-as-Code Output Structure

MANDATE-primary’s distinctive output structure carries fields that the `baseline_specification` schema does not represent. A schematic comparison:

Field	<code>baseline_specification</code>	<code>mandate-as-code</code>
<code>mission_intent</code>	free-text	inside anchor
<code>minimum</code> (cardinality)	array of N	single object
<code>target</code> (cardinality)	array of N	single object
<code>constraints</code>	array	array (often empty)
<code>suspected_gaps</code>	array	<i>not in anchor</i>
<code>courses_of_action</code>	<i>absent</i>	array (always length 1)
<code>recommendation</code>	<i>absent</i>	rationale + <code>primary_coa</code>
<code>trace</code>	<i>absent</i>	cryptographic hash chain
<code>gap_reports</code>	<i>absent</i>	array (currently always [])
<code>registry_reference</code>	<i>absent</i>	tool registry resolution

The baseline cross-comparison’s “9 minimum entries vs 1” framing captures one dimension. The full structural picture is that MANDATE produces additional output components (COA structure, trace chain, recommendation rationale) that baselines do not. Whether those additional components compensate for the anchor-density gap is what Phase 8 grading will quantify on each of the five pre-registered outcomes; the structural comparison here documents the raw counts the grading consumes.

5.4 Phase 8 Three-Judge Grading (Compute-Constrained Sampled Run)

Numerical results below are from the v1 D-08 sampled-grading run that completed 2026-06-22 with a PROTOCOL_LOCK §8 HALT. Per Deviation D-09, the v2 pivot supersedes these numbers; the v1 results are preserved as the schema-mismatch baseline.

Sampled grading protocol. Phase 8 grading operates on a stratified random sample of N=700 anonymized RunRecords (100 per system across the seven systems: `mandate_primary`, `baseline_1`–`baseline_6`). Sample drawn with deterministic seed 20260618; manifest at `08_grading/sample_manifest.jsonl`. The 9,000 anonymized RunRecords remain on disk as canonical artifacts; only the grading layer is sampled. Sample-size justification (statistical power) is in Appendix X. Cost justification is in Deviation D-08.

Outcomes graded.

- O1 anchor completeness: ensemble score per outcome on the locked rubric.
- O2a/O2b gap detection: precision and recall against ground-truth `suspected_gaps`.
- O3 fabrication rate: per-record count of fabricated thresholds, sources, or constraints.
- O4 schema validity: parseable / schema-compliant / consumable-without-repair.
- O5 adversarial resistance: deferred to the perturbation-suite grading run if PI scope permits.

Judge ensemble (revised). Three judges, one per vendor family, per Deviation D-08:

- **Anthropic:** `claude-sonnet-4-6` (was Opus 4.6; swapped for cost; documented self-grading-bias caveat on the B1 column applies — B1 baseline also uses Sonnet).
- **OpenAI:** `gpt-4o-2024-11-20` (unchanged).
- **Google:** `gemini-2.5-pro` with 8192 thinking-token budget (unchanged).

Inter-rater reliability. 10% of the main sample (N=70 records) is double-graded across all three judges for pairwise Cohen’s κ computation. Reduced from the pre-registered 20% under Deviation D-08. Under the SME-skip caveat (Section 2.4), κ measures judge-vs-judge agreement on a Claude-Opus-generated answer key, not judge-vs-SME-truth. The 0.40 halt threshold (PROTOCOL_LOCK §8) remains binding; results below 0.40 trigger a methodological review prior to deposit publication.

v1 grading completed 2026-06-22 — HALT under PROTOCOL_LOCK §8. Final main-pass results on the N=700 D-08 sample:

Metric	Value
Main records graded	700 / 700
Incompletes	0
Double-grade pass1	70 / 70
Double-grade pass2	70 / 70
Total Phase 8 API spend (13f)	\$58.82
Cumulative Phase 8 spend (incl. 13d sunk)	~\$559
Main-pass min pairwise κ	0.296 (Sonnet-Gemini on mission_intent_match)
PROTOCOL_LOCK §8 threshold	0.40
irr.json halt flag	true

Main-pass pairwise κ :

Outcome	GPT-4o $\times$ Sonnet	GPT-4o $\times$ Gemini	Sonnet $\times$ Gemini
mission_intent	0.331	0.709	0.296
gap_class	0.677	0.449	0.713

Per-outcome Krippendorff’s α (pass1 double-grade) splits the ensemble into reliable and unreliable outcomes:

Outcome	α
gap_classification	0.701
minimum_coverage	0.612
target_coverage	0.541
constraint_coverage	0.529
mission_intent_match	0.299
fabrication_count	0.216
trace_completeness	0.194

Objective outcomes (gap_classification, minimum_coverage, coverages) cleared or approached the threshold; subjective outcomes (mission_intent_match, trace_completeness, fabrication_count) fell below it.

Per-system ensemble summary (100 records per system):

System	Mission	Min cov	Tgt cov	Con cov	Gap TP	Sonnet Δ
baseline_1	1.00	0.88	0.68	0.84	100	+0.04
baseline_2	0.26	0.27	0.17	0.27	5	+0.06
baseline_3	1.00	0.96	0.70	0.91	100	+0.08
baseline_4	1.00	0.89	0.66	0.84	100	+0.05
baseline_5	1.00	0.85	0.64	0.85	100	+0.07
baseline_6	1.00	0.88	0.64	0.83	100	+0.04
mandate_primary	0.94	0.18	0.07	0.01	0	−0.30

Interpretation under the v1 rubric — and why v2 supersedes it. A naive reading of the table above is “MANDATE-primary underperforms five of six baselines on every coverage outcome by 5–10 $\times$.” That reading is incorrect under the schema-mismatch finding documented as Deviation D-09 (Section 17) and Finding 6 (Section 4.6). The v1 rubric scored minimum/target/constraint coverage by structural match against an array-of-dimensions ground truth shape. Canonical MANDATE emits these as free-form objects, which the v1 rubric was not authored to consume. The

0.30 Sonnet delta against `mandate_primary` (versus +0.04 to +0.08 for every baseline) is a textbook signature of shape-bias against the canonical MANDATE schema: the same model family that scored everything else higher than its peers scored MANDATE-primary 30 points lower than its peers. The `PROTOCOL_LOCK` κ halt on `mission_intent_match` is itself a methodological finding (Finding 6).

What the v2 pivot replaces (Framing 2 — full pre-registration coverage). Per Deviations D-09 / D-10 / D-11, v2 produces:

- **Cond-X regrade (full).** All 1,500 v1 MANDATE-primary main-matrix records re-graded under the v2 shape-neutral rubric. The per-record delta ($v2 - v1$) quantifies the schema-mismatch penalty as a publishable methodology finding under full statistical power.
- **Cond-A grade (full).** All 1,500 records from canonical MLT v1.0.0rc1 MANDATE on pre-extracted structured `MissionInput`, graded under v2 rubric.
- **Cond-B grade (full).** All 1,500 records from canonical MLT MANDATE with LLM-augmented Interpreter end-to-end on raw text, graded under v2 rubric.
- **Baselines re-graded (full).** All 7,500 baseline records (B1–B6) re-graded under the v2 rubric for apples-to-apples cross-system comparison.
- **Pre-registered judge ensemble restored.** Claude Opus 4.6 as the Anthropic judge (D-10), GPT-4o, Gemini 2.5 Pro. 20% IRR sample. `PROTOCOL_LOCK` §8 $\kappa \geq 0.40$ binding.
- **O5 measured.** 24,500 perturbation records produced and graded (D-11). Adversarial compliance computed on the 3,500-record prompt-injection stratum.

Total v2 cost under D-10 + D-11 (full pre-registration coverage, no sampling, no judge substitutions, no deferred outcomes save the ablations that depend on upstream MANDATE source changes): ~\$15,256. Section 5.4 numerical claims about MANDATE performance are deferred to the v2 deposit; the v1 numbers above are preserved as the schema-mismatch baseline that motivates the v2 pivot.

6 Pre-Phase-6 Demo Evidence

The MANDATE-primary evaluation built up from three real-world demo scenarios before committing to the Phase 6 main matrix. Each demo scenario was run through three apparatus variants. The cross-variant comparison is the demo-era evidence that the Interpreter and Binding content-tripwires are driven by chunk shape rather than the user-facing request paragraph (which is identical across variants for a given scenario).

6.1 The Three Scenarios

Volt Typhoon (security ops). CISO request for defense capability against the PRC state-sponsored intrusion campaign documented in CISA AA23-144A and AA24-038A. 12 source files including the CISA joint advisories (PDF), MITRE ATT&CK G1017 group page (HTML), Microsoft Threat Intelligence original disclosure blog post (HTML), DOJ KV Botnet press release (HTML), FBI search warrant affidavit 24-5018 (PDF), Wikipedia coverage, NIST SP 800-53/61. Mixed PDF + HTML.

CrowdStrike Outage (security ops). CIO request explicitly demanding “three distinct strategic options for endpoint security architecture” after the 19 July 2024 Falcon Sensor outage. 13 source files including the CrowdStrike Channel File 291 RCA (PDF), Microsoft recovery guidance (HTML), CISA advisory (HTML), SEC 8-K filings from Delta and United (HTML), House Homeland Security Committee testimony (HTML), NIST SP 800-218 deltas (**DOCX**), NIST CSF v1.1 presentation (**PPTX**), Intro to Framework Roadmap (PPTX). Multi-format ingest: PDF + HTML + DOCX + PPTX.

Silicon Valley Bank Collapse (financial reporting). CFO request explicitly demanding “three distinct strategic options for ERM and treasury overhaul” after the March 2023 Silicon Valley Bank failure. 11 source files including the Federal Reserve SVB Review April 2023 (PDF), Senate Banking Becker testimony (PDF), Federal Reserve and FDIC press releases (HTML), SEC EDGAR 10-K filings (HTML), NIST SP 800-218 / 800-53 deltas (DOCX), NIST CSF v1.1 and Roadmap presentations (PPTX), Wikipedia coverage.

6.2 Three Apparatus Variants Per Scenario

- **output (deterministic apparatus, no LLM):** The MANDATE pipeline runs without LLM role calls; deterministic rules produce a baseline mandate-as-code artifact for sanity. No Ollama dependency.
- **output_ollama (Ollama mode, mcp web_fetch chunks):** The full six-role Ollama pipeline against scenario-specific RAG indices built from text extracted by `mcp web_fetch` text-mode rendering.
- **output_ollama_from_binaries (Ollama mode, binary extraction chunks):** The same six-role pipeline against scenario-specific RAG indices built from the original PDF/DOCX/PPTX binaries extracted by the apparatus’s production `pypdf`, `python-docx`, and `python-pptx` extractors.

The cross-variant comparison isolates the chunk-content variation: same scenario task text, same six fine-tuned models, same per-role temperatures, same prompts; only the chunks fed to the Procedure-role retriever differ between variants.

6.3 Cross-Variant Demo Results

Scenario	Variant	OK	Wall (s)	LLM fb.	n_{COAs}	Fb. roles
volt_typhoon	output	T	0.0	F	1	—
volt_typhoon	ollama	T	238.5	F	1	—
volt_typhoon	ollama_binaries	T	166.7	F	1	—
crowdstrike	output	T	0.0	F	1	—
crowdstrike	ollama	T	212.0	F	1	—
crowdstrike	ollama_binaries	T	172.0	F	1	—
svb_collapse	output	T	0.0	F	1	—
svb_collapse	ollama	T	195.8	F	1	—
svb_collapse	ollama_binaries	T	215.1	T	1	Binding

All nine demo runs produce $n_{COAs} = 1$, reaffirming the Decomposition single-COA prior at the demo level. The only LLM fallback in the demo work is the SVB from-binaries Binding refusal (Section 8.4 below).

6.4 The Interpreter Mode Flip Cross-Variant

The verbatim `anchor.minimum.description` per scenario per variant. This is the demo-era smoking gun for the content-tripwire mode flip: the request paragraph is identical across variants; the chunks fed to the upstream Procedure retriever differ; the Interpreter output mode flips accordingly.

Volt Typhoon (stable: `det-prefix` in all three variants).

```
output:
  Minimally satisfy: Team, this is the CISO speaking. CISA
  has now published two binding-direction advisories
  (AA23-144A and AA24-038A) ...

output_ollama:
  Minimally satisfy: Team, this is the CISO speaking. CISA
  has now published two binding-direction advisories
  (AA23-144A and AA24-038A) ...

output_ollama_from_binaries:
  Minimally satisfy: Team, this is the CISO speaking. CISA
  has now published two binding-direction advisories
  (AA23-144A and AA24-038A) ...
```

CrowdStrike (flips: `web_fetch` → `clean`, others → `det-prefix`).

```
output:
  Minimally satisfy: Team, this is the CIO. As we discussed
  at the executive briefing, the board has formally
  requested a strategy memo follo...

output_ollama (web_fetch chunks):
  Three distinct strategic options for endpoint security
  architecture with (a) operational and capex
  implications, (b) residual risk profile after one year,
  (c) vendor-relationship and contractual changes, and
  (d) implementation timeline.

output_ollama_from_binaries (binary extraction chunks):
  Minimally satisfy: Team, this is the CIO. As we
  discussed at the executive briefing, the board has
  formally requested a strategy memo follo...
```

SVB (flips in opposite direction: `from-binaries` → `clean`, others → `det-prefix`).

```
output:
  Minimally satisfy: Team, this is the CFO. The audit
  committee has asked for a strategy memo following the
  Silicon Valley Bank collapse of M...

output_ollama (web_fetch chunks):
  Minimally satisfy: Team, this is the CFO. The audit
  committee has asked for a strategy memo following the
  Silicon Valley Bank collapse of M...

output_ollama_from_binaries (binary extraction chunks):
  Three distinct strategic options with operational/capex
  implications, residual risk profiles modeled against
  SVB-class deposit outflow, contractual/personnel
  changes, and implementation timelines aligned with next
```

CrowdStrike and SVB *flip in opposite directions* between the web_fetch and from-binaries variants. Same scenario; same request paragraph; same fine-tuned Interpreter model. The clean-distillation output appears on CrowdStrike under web_fetch chunks and on SVB under from-binaries chunks. The deterministic-prefix output appears on CrowdStrike under from-binaries chunks and on SVB under web_fetch chunks. This is the demo-era smoking gun for the finding documented at Phase 6 scale (Section 4.2): the Interpreter mode is selected by chunk content, not by request paragraph.

6.5 The Binding Refusal Demo Provenance

The SVB from-binaries run triggered the Binding refusal (last row of Section 8.3 table). The dedicated Binding-refusal diagnostic captured three consecutive direct-Ollama calls against this failing pipeline state; verbatim quotes are in Section 4.4 of this document. The diagnostic established the structured-refusal pattern that Phase 6 measured at scale (244 of 1500 records).

6.6 Demo Findings Validated at Phase 6 Scale

The five demo-era findings and their Phase 6 replication status:

Demo finding	Phase 6 replication	Phase 6 sample
Decomposition single-COA prior	Yes (1480/1480)	MP main + holdout
Interpreter mode flip	Yes (per-domain rates)	MP main (1180 ok)
Validator gap-instability	Yes (14-19% range)	MP main (1180 ok)
Binding structured refusal	Yes (244/1500, FIN low)	MP main + holdout
Intake content-tripwire	NEW at Phase 6 (20/1200)	Two SEC tasks

Four of five demo findings replicated cleanly at scale. The fifth (Intake content-tripwire) emerged only at Phase 6 scale because the affected phrase “Here’s the constraint:” appears in only 2 of 120 main-corpus tasks, both outside the 6-task pilot smoke (Section 10).

7 Pilot Smoke Test

The pilot smoke pass (2026-06-05) ran the full 7-system $\times$ 6-pilot-task matrix at 1 run each as a smoke test before committing to the multi-day Phase 6 main matrix. The 42 records confirmed the apparatus runs end-to-end against the actual ground truth (the pilot scaffolds) and not just against the demo scenarios. Per-system results:

System	Records	OK	Schema-valid	LLM fallback	Binding refusal	Mean wall-clock (s)
mandate_primary	6	6	n/a	0	0	113.0
baseline_1	6	6	6	0	0	38.1
baseline_2	6	6	6	0	0	4.7
baseline_3	6	6	0	0	0	64.0
baseline_4	6	6	6	0	0	65.7
baseline_5	6	6	6	0	0	46.4
baseline_6	6	6	6	0	0	38.2

The pilot smoke established two patterns that Phase 6 confirmed at scale. First, B3 already produced 0/6 schema-valid records at the calibration tier; the same failure mode (ReAct’s structurally-flat JSON) replicated at 0/1205 at Phase 6 (Section 5.2). Second, MANDATE-primary produced 6/6 ok records at ~ 113 seconds mean wall-clock; Phase 6 main matrix MANDATE-primary records replicate this latency at ~ 105 seconds median (Section 3.3).

Schema-valid n/a for MANDATE-primary. The `schema_valid` field applies to baseline outputs (which use the `baseline_specification` schema). MANDATE-primary outputs use the `mandate-as-code` schema; the field is unset on those records by apparatus convention. The mandate-as-code artifact is validated through a different mechanism documented in the apparatus README.

8 Baseline Calibration Detail

the B4-B6 calibration (2026-06-04), the B1-B3 calibration initial (2026-06-04), and the B3 completion under corrected escalation rule, 2026-06-04) calibrated each baseline on the 6 calibration tasks (2 per domain) to confirm end-to-end runs against live API keys before committing to the Phase 6 spend. Calibration totals: 36 RunRecords.

Baseline	Records	OK	Schema-valid	Rate
baseline_1 single-prompt Claude	6	6	6	100.0%
baseline_2 single-prompt GPT-4o	6	6	5	83.3%
baseline_3 ReAct Claude	6	6	0	0.0%
baseline_4 AutoGen	6	6	6	100.0%
baseline_5 CrewAI	6	6	6	100.0%
baseline_6 LangGraph	6	6	6	100.0%

B2 single bare-number threshold. The one B2 calibration schema-validity failure was on TASK-CAL-INT-001: the intelligence-collection task’s anchor minimum’s `threshold` field was emitted as the literal number 25 instead of the string "25". The schema requires `string` or `null` for the threshold field. This is the same B2 calibration finding that replicated at 5/6 schema-valid (one failure on the same INT task) at the calibration stage; at Phase 6 main matrix scale B2 produces 1205/1206 schema-valid records (99.92%). The B2 calibration’s bare-number-threshold case generalizes: GPT-4o emits structurally-correct anchors with occasional type coercion failures on specific dimension values.

B3 ReAct 0/6 replicates to 0/1205. B3’s calibration-tier schema-validity rate (0/6) replicated exactly at Phase 6 scale (0/1205). The ReAct framework on Claude converges on structurally-flat JSON patterns. Three distinct failure shapes were observed during calibration inspection: (a) the entire `minimum`, `constraints`, and `suspected_gaps` fields emitted as keyed objects instead of arrays-of-objects; (b) `constraints` emitted as an array of plain strings; (c) extra prose around the JSON. None of these match the locked `baseline_specification` schema. The Phase 6 replication at $200\times$ scale confirms the framework’s structural failure is stable, not noise.

9 Execution Chronology

The v1 empirical work was executed in three sweeps. The first sweep (2026-06-03 to 2026-06-05) covered corpus authoring, baseline calibration across all six baselines, anchor scaffold generation for the 150-task main and hold-out corpora, generation of the 350-perturbation suite, and the three from-binaries demo replays. The second sweep (2026-06-05 to 2026-06-13) executed the Phase 6 main matrix on MANDATE-primary and the six baselines plus the hold-out generalization run. The third sweep (2026-06-16 to 2026-06-22) executed Phase 8 grading on the 700-record D-08 sample. The post-Phase-8 analysis sweep (2026-06-23) identified the schema-mismatch issue and initiated the v2 pivot.

Across these sweeps, 39 separately-tracked execution segments were run to support apparatus iteration and incremental correction. The pattern across the chronology was substantive discovery followed by apparatus correction followed by re-fire under the corrected rule — a pattern that recurred for the corpus resolver, the scaffolder token budget, the watchdog rule, the grader checkpoint flush semantics, and the judge retry layer. Each correction has a companion regression test and a committed patch on the project main branch. The full segment-by-segment execution log is preserved in the project’s `handoffs/` directory and bundled in the replication package; this supplement summarizes substantive outcomes rather than reproducing the operational log.

10 Apparatus Engineering Provenance

This section documents the apparatus patches applied during Phase 6 execution. None of these patches modify the MANDATE-primary v1 frozen tag (`mandate-eval-primary-2026q2-v1`, commit `4f8af83`); all patches are confined to the wrapper apparatus that orchestrates the evaluation.

10.1 Scaffolder Token Budget Patch

Initial PROMPTS Section 2 anchor scaffolding under `max_tokens=2048` produced truncated JSON that failed balanced-brace parsing in all six pilot scaffolds (0/6 `parse_ok`). Patched the `AnchorScaffolder` default to `max_tokens=4096` and exposed `-max-tokens` on the CLI. The `raw_json` field was added to `ScaffoldedAnchor.to_dict` for failure diagnosis. Post-patch: 156/156 `parse_ok` on pilot + main + hold-out scaffolds.

Regression test count: 73 corpus tests passing post-patch.

10.2 Apparatus Resume Support

The `apparatus.run run-system` CLI had no resume support. A re-fire on a partial output directory re-executed every (task, run) tuple and overwrote committed records. Patched `run_matrix` to accept `skip_existing=True` and exposed the `-skip-existing` CLI flag. When set, existing `<run_id>.json` files are loaded into the ledger but `system.run()` is not called for those tuples.

Regression tests: two new tests cover (a) skip-existing does not re-execute existing records and does not rewrite their files (mtime check); (b) skip-existing correctly resumes a partial checkpoint by executing only the missing tuples. All 7 harness tests passing.

10.3 Three-Judge Grading: Double-Grade Sample and Gemini Budget

Double-grade CLI patch: The grader CLI did not invoke `pipe.double_grade` despite `PROTOCOL_LOCK §8` requiring a 20% IRR sample. Patched `cmd_grade` to support `-double-grade-ptc`

and `-double-grade-seed` flags.

Per-instance max_tokens patch: The `Judge.grade()` method hardcoded `max_tokens=2048`, sufficient for GPT-4o and Claude Opus 4.6 but insufficient for Gemini 2.5 Pro (which consumes output tokens for thinking-mode reasoning before emitting the visible response). Patched `Judge.__init__` to accept per-instance `max_tokens` (default 2048); the `judge_gemini_pro` factory passes 8192.

Regression tests: 17 grading tests passing post-patch including double-grade independent-pass coverage and per-instance `max_tokens` coverage.

10.4 Watchdog Refinement and Hold-Out Contamination Correction

Two related watchdog defects were corrected during Phase 6 execution:

Watchdog rule refinement: The original watchdog halted on any record with `wall_clock_ms < 60000`. This conflated Ollama-crash contamination (all six roles fell back to deterministic with `wall_clock` $\approx$ 2 seconds) with legitimate Intake failures (Intake LLM ran for $\approx$ 11 seconds before validation rejected the constraint string; remaining roles never ran). The refined rule halts only when *all* six roles report `llm_used=False` and `wall_clock_ms < 60000`.

Hold-out contamination signature: Even the refined rule missed a 300-record contamination event in the MANDATE-primary hold-out leg, because the apparatus sets `llm_used=True` before attempting the LLM call and `llm_fallback=True` after the call fails. Both flags are True on a contaminated record. The corrected detection signature is *all roles llm_fallback=True*, not *all roles llm_used=False*. The 300 contaminated records were quarantined, Ollama was restarted, and 300 fresh records were produced cleanly (0/300 contamination, 21/300 legitimate Binding refusal).

10.5 Canonical MANDATE Discovery and v1 Drift Confirmation (2026-06-23)

Mid-evaluation analysis following the Phase 8 HALT (Section 5.4) identified that the v1 measurement ran the experimental `AEGIS-eval/src/mandate` fork (71 files, 6,793 LOC) rather than canonical MANDATE. The canonical MANDATE release exists at `/Users/ws01admin/Desktop/MLT-Governance-Stack` as `mlt-stack 1.0.0rc1` (released 2026-06-13). The release ships:

Plane	Modules	LOC
<code>mlt.mandate</code> (specification — six-role pipeline)	28	10,639
<code>mlt.lattice</code> (decision — policy engine + gate)	18	3,209
<code>mlt.trace</code> (evidence chain + verifier)	19	5,218
<code>mlt.core</code> / <code>sdk</code> / <code>envelope</code> / <code>gateway</code> / <code>verify</code> / <code>policies</code> / <code>ui</code>	47	~5,900
Total	116	25,199
Tests collected (pytest)	1,942	—
JSON schemas	19	—

Drift verification. MD5 hashes of all six role implementations under `AEGIS-eval/src/mandate/roles/` differ from canonical `MLT-Governance-Stack/src/mlt/mandate/roles/` for every file (`intake.py`, `interpreter.py`, `decomposition.py`, `procedure.py`, `binding.py`, `validation.py`). The AEGIS-eval Intake schema added experimental `time_limit`, `minimum_outcome`, `target_outcome`, `available_tools`, `risk_tolerance`, and `metadata` properties not present in the canonical version. The canonical `mandate-as-code.schema.json` specifies `anchor.minimum` and `anchor.target` as free-form objects (`type: object`, `minProperties: 1`,

`additionalProperties oneOf [number, string, boolean, object, array]`), confirming the schema-mismatch root cause documented as Finding 6 (Section 4.6).

Realness verification. Three-source convergent verification (`handoffs/MLT_realness_audit_opus.md`):

- **Opus subagent audit:** REAL at 85% confidence. 418 mandate tests passing in 0.8 seconds. Verified hand-written recursive-descent constraint parser with ReDoS mitigation, hash-anchored trace chain with RFC 8785 JCS canonicalization, 4-step validation algorithm with real per-COA scoring math, typed gap taxonomy, Rego/Cedar policy translators (313 + 253 LOC of real translator code plus 729 LOC of translator tests).
- **GPT-4o** independent review: real production-grade.
- **Gemini 2.5 Pro** independent review: real, non-trivial.
- **Direct empirical pipeline run:** produced canonical 3-COA artifact on `tests/examples/normal_mission.json`; also produced single-COA `{description: ...}` shape on five v1 corpus tasks (TASK-MAIN-INT-034 CW threat matrix, FIN-001, FIN-018, INT-003, SEC-014) — identical to v1 records on the same tasks.

Decision to pivot. The canonical-MANDATE-vs-v1-records shape equivalence on the same task inputs is dispositive: the v1 records ARE valid canonical MANDATE outputs. The v1 “MANDATE underperforms baselines” reading conflates rubric-shape mismatch with substantive coverage. v2 (D-09, Section 17) replaces the drifted AEGIS-eval source with canonical MLT v1.0.0rc1, rewrites the rubric to be shape-neutral, and runs three measured conditions (Cond-X regrade + Cond-A pre-extracted + Cond-B LLM-augmented) plus re-graded baselines on the same corpus.

10.6 The v2 Binding-Refusal Candidate Patch (Not Installed for v1)

A candidate patch on branch `feature/binding-refusal-as-gap-sideload` extends the apparatus to detect Binding’s structured refusals and route them to `output.gap_reports` instead of treating them as parse failures. The patch:

1. Adds a `detect_structured_refusal()` helper to `aegis/llm/response_parser.py`.
2. Extends `ResponseParseError` to carry the last raw response and last parsed payload.
3. Modifies `generate_validated_response` to short-circuit on refusal (no retries when the response is a clean refusal).
4. In `BindingRole.execute_with_llm`, catches refusal, runs the deterministic recommendation as a fallback summary, annotates the artifact with `llm_refused_with_error=True` and the verbatim model text, returns cleanly without a fallback flag.
5. In `pipeline.py`, injects a `GapSpec(UNASSESSABLE_RISK, detected_by="Binding")` into `output.gap_reports` carrying the refusal text and a recommended action.

Verification at three levels: unit tests (1448 AEGIS-eval tests passing, including 5 new refusal-specific cases); deterministic stub-adapter replay of the captured refusal payload (short-circuit fires on the first call, payload preserved, gap report emitted, rationale prefixed [`binding refused`]); and negative testing (Volt Typhoon and CrowdStrike v2 runs do not over-fire). The patch is *not* installed for v1 measurement; v1 measures the unpatched behavior.

11 Perturbation Suite

The 350-perturbation suite was generated from the frozen main corpus by the perturbation-suite generation and frozen at the `perturbation_freeze_v1` tag (commit `068fe94`). The suite carries seven canonical perturbation types at 50 perturbations each. Each perturbation references a base task ID and applies a specific perturbation function to the task text.

Perturbation type	Count
<code>ambiguity_injection</code>	50
<code>contradictory_constraints</code>	50
<code>length_perturbation</code>	50
<code>missing_required_field</code>	50
<code>out_of_distribution_input</code>	50
<code>prompt_injection</code>	50
<code>surface_noise</code>	50
Total	350

Record schema. Each perturbation record carries `perturbation_id`, `base_task_id` (the source main-corpus task), `perturbation_type` (one of seven), `sub_type` (category within a type — e.g., for `prompt_injection`: `direct_override`, `context_break`, `role_swap`), `domain`, `request_text` (the perturbed task text the system under test sees), `generation_ms`, `input_tokens`, `output_tokens`, `model`, and an `internal_note` describing the perturbation applied.

Phase 6 status. The perturbation suite is part of the frozen deposit but the system runs against it (the planned O5 adversarial-resistance evaluation) are deferred per the PI scope decision. The suite provenance and structure are documented so a follow-up study can evaluate O5 against it without regenerating the suite.

Why seven types. The seven types span complementary adversarial dimensions: `ambiguity_injection` adds genuinely ambiguous references; `contradictory_constraints` introduces logical contradictions between requirements; `length_perturbation` tests robustness to abnormal length (very short or very long requests); `missing_required_field` strips a critical specification element; `out_of_distribution_input` substitutes domain-irrelevant text; `prompt_injection` adds adversarial instructions inside the request; `surface_noise` adds whitespace, typos, formatting glitches. Together they probe different specification failure modes a real-world system might encounter.

12 Corpus Candidate Pool Audit

The deposit’s selected tasks (120 main + 30 hold-out + 6 pilot) are a subset of larger candidate pools produced by the source-conditioned generation pipeline. The unused residue is a v2 evaluation resource.

12.1 Main Corpus

Main corpus state	Count
Total candidates generated	262
Selected for Phase 6	120
Residue (available for v2)	142

Residue by domain: financial_reporting 49, intelligence_collection_tasking 46, security_operations_reporting 47.

Residue by category: full_specification 46, gap_triggering 47, stretch_case 49.

The residue’s distribution by domain and category is roughly balanced. The DEVIATIONS.md (2026-06-10 Intake content-tripwire entry) cites this 142-candidate residue as the v2-candidate-replacement source for TASK-MAIN-SEC-038 and TASK-MAIN-SEC-040 (the Intake-tripwire tasks). A future evaluation could swap those tasks with alternative SEC stretch_case candidates from the 49-strong SEC residue pool to test whether the substantive findings hold against a different task set.

12.2 Hold-Out Corpus

Hold-out corpus state	Count
Total candidates generated	44
Selected for Phase 6	30
Residue (available for v2)	14

The 14-candidate residue covers all three categories (full_specification 4, gap_triggering 5, stretch_case 5) and can be used for v2 hold-out expansion if a larger generalization sample is needed.

12.3 Pilot Corpus

Pilot corpus state	Count
Total candidates generated	183
Selected for pilot scaffolds	6
Residue (available for v2)	177

The pilot pool is much larger than the selected pilot subset (183 vs 6). A future v2 evaluation that wants a larger pilot sample (for example, 30 tasks across the three main-corpus domains for a faster smoke test before the full 120-task main run) has 177 unused candidates available.

13 v2 Candidate Patch: Full Verification Trail

The v2 candidate Binding-refusal patch is on branch `feature/binding-refusal-as-gap-sideload`. The patch routes `mandate-binding`’s structured `{"error": "..."}` refusals into `output.gap_reports` rather than discarding them after 3 parse retries. Verification was performed at three levels.

13.1 Level 1: Unit Tests

Total tests passing on the v2 branch: 1448 AEGIS-eval tests (1443 baseline + 5 new refusal-specific cases).

New tests in `tests/test_binding_refusal.py`:

- `test_detect_structured_refusal_returns_text` — verifies the helper returns the error text on a clean `{error: <str>}` payload.
- `test_detect_structured_refusal_rejects_extra_keys` — verifies the helper rejects payloads with extra keys, non-string error values, empty strings, or non-dict shapes.
- `test_generate_validated_response_short_circuits_on_refusal` — verifies the parse loop does NOT consume the retry budget when the response is a refusal; adapter called exactly once.
- `test_binding_role_handles_structured_refusal` — verifies `BindingRole.execute_with_llm` catches the refusal and annotates the result with `llm_refused_with_error=True`.
- `test_binding_role_falls_back_on_malformed_json` — verifies `BindingRole.execute_with_llm` re-raises on malformed JSON so the existing pipeline-level fallback still runs.

13.2 Level 2: Deterministic Stub-Adapter Replay

The Binding refusal diagnostic captured three direct-Ollama refusal payloads from the SVB from-binaries scenario; all three were clean JSON with the single `error` field. The stub-adapter replay feeds the first captured payload through the patched apparatus with a stub adapter that returns the captured text instead of calling Ollama. The verification asserts:

- `generate_validated_response` short-circuit fires on the first call (zero retries).
- `last_parsed_payload` is preserved on the raised `ResponseParseError`.
- Adapter called exactly once on refusal (not 3 times as the original behavior).
- Pipeline produces a `GapSpec(UNASSESSABLE_RISK, detected_by="Binding")` carrying the verbatim model error text.
- Recommendation rationale is prefixed with `[binding refused]`.

13.3 Level 3: Negative Testing

Volt Typhoon and CrowdStrike full-pipeline v2 sanity runs confirmed the patch does not over-fire on non-refusal scenarios:

- Volt Typhoon v2: `any_llm_fallback=False`; zero Binding-attributed gap reports.
- CrowdStrike v2: `any_llm_fallback=False`; zero Binding-attributed gap reports.

These are the same scenarios from the demo work (Section 8) re-run against the patched apparatus. Since their original demo runs did not refuse, the v2 patch should not emit refusal gap reports for them either. Confirmation that the refusal predicate is narrow.

13.4 Probabilistic Refusal at Repeated-Run Level

The repeated-run lane ran SVB v2 five times on the same task against the patched apparatus. Result: 0 of 5 runs refused (the original demo SVB from-binaries run had refused at demo time; the repeated-run lane measured the probabilistic rate). Combined across the various demo and stub verification runs, the observed full-pipeline SVB refusal rate is approximately 1 of 7 ($\approx 14\%$); the direct-Ollama refusal rate against the captured prompt at diagnostic time was 3 of 3 (100%). The difference is the upstream pipeline's stochastic outputs gating whether the Binding role sees a contradictory anchor in any given run.

13.5 Patch Surface (Files Modified)

- `src/aegis/llm/response_parser.py`: added `detect_structured_refusal()` helper; extended `ResponseParseError` with `last_raw_response` and `last_parsed_payload`.
- `src/mandate/llm_support.py`: `generate_validated_response` short-circuits on detected refusal.
- `src/mandate/roles/binding.py`: `BindingRole.execute_with_llm` catches the refusal, runs deterministic recommendation as fallback summary, annotates artifact.
- `src/mandate/pipeline.py`: injects `GapSpec(UNASSESSABLE_RISK, detected_by="Binding")` when `llm_refused_with_error=True`.
- `tests/test_binding_refusal.py`: new 5-case test file.

13.6 Why v1 Does Not Install the Patch

PROTOCOL_LOCK §13 binds the formal Phase 6 study to the v1 frozen tag `mandate-eval-primary-2026q2-v1` (commit `4f8af83`). The v2 patch is a candidate evaluation question whose answer is what a future v2 study would measure. For the v1 study, the Binding refusal failure path is the apparatus behavior, and the 244/1500 refusal records are valid Phase 6 measurement data: the model correctly refuses, the apparatus discards the explanation, and the trade-off is documented.

14 Realism Audit Machinery: Forward Compatibility

The realism audit CLI infrastructure (`apparatus.corpus.cli realism-form`, `apparatus.corpus.cli realism-aggregate`) is built and tested, but unused in the v1 evaluation per the SME-skip deviation (DEVIATIONS.md 2026-06-04).

The CLI.

- `realism-form -selection <selection.json> -pool <candidates.jsonl> -rater-id <id>` produces a per-rater CSV template for an SME to fill in with realism ratings for each candidate.
- `realism-aggregate` consumes the filled-in CSVs across multiple raters and produces the audit report including Krippendorff's α inter-rater reliability and the halt list (candidates flagged below the threshold).

Forward-compatibility statement. For any future v2 evaluation that adds SME involvement, the existing deposit artifacts feed cleanly into the realism audit:

1. The 156 PROMPTS Section 2 scaffolds under `04_ground_truth/{pilot,main,holdout}_scaffolds/` become candidate anchors for SME accept/edit/reject review.
2. The realism audit CSV produced by `realism-form` can be sent to SMEs for the 120 main-corpus tasks.
3. `realism-aggregate` consumes the filled CSVs to compute α and the halt list, gating a `corpus_freeze_v2_with_sme` tag.
4. The existing Phase 6 RunRecords at `outputs_freeze_v1_1` can be re-graded against the SME-revised ground truth without regenerating the run-time data.

The CC BY 4.0 license on this deposit permits independent re-grading work without permission. Independent SME teams can produce a v2 ground truth on top of this v1 deposit and report differences in O1-O4 outcomes attributable to SME revision.

15 Per-Phase Cost Log

The full Phase 6 evaluation, including Phase 8 grading (currently in flight at the time of initial writing), totaled approximately \$2,000-\$2,500 in API spend. The MANDATE-primary main matrix and hold-out runs use local Ollama and incur no API cost; the cost falls into corpus generation, baseline runs, and three-judge grading.

Phase / Activity	Cost (USD)
B4–B6 calibration	\$0.87
B1–B3 calibration	\$1.00
B3 completion under corrected escalation rule	\$0.50
Pilot scaffolds (6 tasks)	\$1.00
Hold-out corpus authoring (30 tasks)	\$3.00
Main anchor scaffolds (120 tasks)	\$12.00
Perturbation suite generation	\$3.09
Pilot smoke (7 systems $\times$ 6 tasks)	\$1.51
Phase 6 MP main matrix (Ollama-only)	\$0.00
Phase 6 baselines + hold-out + freeze tags	\$339.03
Hold-out anchor scaffolds (30 tasks)	\$4.00
MP hold-out regeneration (Ollama-only)	\$0.00
Subtotal (pre-Phase 8)	\$366.00
Phase 8 grading killed at 25h (no checkpoints)	~\$500 (est.)
Phase 8 grading completion (D-08 revised plan)	\$58.82 (attested)
v1 close (Phase 8 HALT verdict)	~\$925
v2 Stage 1 apparatus plumbing (operator time)	\$0
v2 Stage 2 pilots (5 tasks $\times$ 2 conditions)	~\$1
v2 Stage 3 Cond-A 1500-record run	~\$30
v2 Stage 3 Cond-B 1500-record run	~\$175
v2 Stage 4 grading (12,000 records, Opus, 20% IRR; D-10)	~\$7,700
v2 Stage 6 O5 perturbation runs + grading (D-11)	~\$7,350
Total v2 deposit-ready (D-10 + D-11, full pre-registration)	~\$16,181

The Phase 8 sunk-cost incident. The initial Phase 8 grading run executed for 25+ hours with zero on-disk artifacts because the original grading pipeline accumulated 9000 records in memory and only flushed via `pipe.save()` after the full main pass completed. The process was killed intentionally; the corrected grader resumed with per-record checkpointing patched into the pipeline plus 3-way judge concurrency. The 13d API spend is best-estimate without on-disk evidence (provider dashboards consulted at kill time).

Computational compute (separate from API spend). Phase 6 main matrix MANDATE-primary required approximately 154 hours of Ollama serial compute on the eval host (Apple Silicon Mac, six 32B-parameter mandate-* role models). Phase 6 hold-out required approximately 31 additional hours (300 records). The compute is a one-time eval-host expense (electricity + thermal stress on the laptop) rather than a recurring cloud spend.

16 Deviation Log

The full chronological deviation log is at `code/00_preregistration/DEVIATIONS.md`. The principal deviations:

2026-06-04: SME-skip. The pre-registered SME realism audit and SME anchor-review workflow were skipped. The PROMPTS Section 2 anchor scaffolds produced by Claude Opus 4.6 serve as ground truth verbatim. Cohen’s kappa under SME-skip measures inter-judge agreement on a Claude-Opus-generated answer key, not inter-judge recovery of SME judgment. The realism question is an acknowledged open caveat.

2026-06-04: Ablation variants deferred. Five of seven pre-registered ablation variants (A1, A2, A4, A6, A7) require upstream MANDATE source changes and are deferred to the upstream MANDATE team’s 2-3 engineer-week timeline. A3 (no gap-report) and A5 (no Success Registry) are configuration switches and run from the apparatus today. The ablation analysis in the main paper is restricted to A3 and A5.

2026-06-10: Watchdog refinement. The original contamination detection rule (halt on `wall_clock_ms < 60000`) over-quarantined legitimate single-role failures. Refined per Section 6.4.

2026-06-13: Hold-out contamination corrected. 300 contaminated MANDATE-primary hold-out records (Ollama outage during the Phase 6 baselines/hold-out generation) were quarantined and regenerated cleanly. `outputs_freeze_v1` preserved as historical record; `outputs_freeze_v1_1` cut at the corrected commit.

2026-06-16: Intake content-tripwire characterized as finding. 20 of 1200 main-matrix runs failed at the Intake role on two tasks carrying the natural-language phrase “Here’s the constraint:”. The failures are recorded as Phase 6 measurement data; the underlying behavior is documented as a v2 candidate apparatus or training-data mitigation question.

Perturbations (O5) deferred. The 350-perturbation suite is generated and frozen at `perturbation_freeze_v1`, but the system runs against it are deferred pending PI scope decision. The O5 adversarial resistance outcome is therefore not measured in this evaluation.

2026-06-17: Phase 8 grading restart with checkpointing patch. The initial Phase 8 grading run executed for 25+ hours with zero on-disk artifacts because the original grading pipeline accumulated all 9000 records in memory and only flushed via `pipe.save()` at the end of the main pass. The process was killed intentionally; the grader was patched to write per-record checkpoints to `08_grading/by_record/<anon_id>.json` immediately and to support `-skip-existing` resume + `-max-workers` 3-way concurrency. The corrected grader resumed under the patched implementation. The initial-run API spend is best-estimate ~\$500 (no on-disk evidence; estimated from provider dashboards at kill time) and is sunk.

16.1 Full DEVIATIONS.md Chronology (Expanded)

The full chronological deviation log carries five entries through 2026-06-17:

2026-06-04 — SME realism audit skipped; ground truth = Claude Opus 4.6 anchor scaffolds direct. PI decision; rationale: focus on apparatus end-to-end proof; forward-compatible with v2 SME re-evaluation. Affects every freeze tag downstream of corpus selection. Cohen’s kappa under SME-skip measures judge-judge agreement on Claude-Opus-generated answer key, not judge-SME agreement.

2026-06-10 — Phase 6 watchdog refinement: ok=False records are data, not contamination. The original watchdog (`wall_clock < 60000`) over-quarantined legitimate Intake failures. Refined to require both fast wall-clock AND all-roles `llm_used=False`. The 2 quarantined SEC-038 records from the Intake content-tripwire failures were restored from quarantine to the canonical output directory. Phase 6 O4 measurement integrity improved.

2026-06-10 — Substantive finding: Intake role content-tripwire on natural-language constraint mentions. 20 of 1200 main-matrix runs failed at the Intake role on two tasks (TASK-MAIN-SEC-038, TASK-MAIN-SEC-040) carrying the phrase “Here’s the constraint:”. The phrase is the trigger; the source document (NIST SP 800-115) is not — 6 of 8 NIST SP 800-115-derived tasks have 0/10 Intake failure rate. Three potential v2 mitigations documented; v1 study runs with the failure mode observed.

2026-06-13 — Hold-out contamination: 300 records invalid, regenerated cleanly. The Phase 6 MANDATE-primary hold-out leg produced 300 records where every role on every record reported `llm_fallback=True` with the reason “Ollama backend failed after 3 attempt(s): Ollama connection error: [Errno 61] Connection refused”. The corrected contamination signature is “all roles `llm_fallback=True`” (not “all roles `llm_used=False`” as previously checked). The 300 contaminated records were quarantined and regenerated cleanly; `outputs_freeze_v1` preserved as historical record; `outputs_freeze_v1_1` cut at the corrected commit.

2026-06-17 — Phase 8 grading restart with checkpointing patch. See the paragraph above. The grader pipeline now writes per-record checkpoints immediately and supports resume + concurrency. Four regression tests guard the patched behavior.

2026-06-18 — D-08: Compute-constrained sampled grading; Sonnet judge substitution; reduced IRR sample. The pre-registered protocol specified full-coverage grading of all anonymized RunRecords (N=9,000 after the `v1_1` freeze) by a three-judge ensemble (Claude Opus

4.6, GPT-4o, Gemini 2.5 Pro) with 20% double-grading for inter-rater reliability (IRR). At full coverage and with the pre-registered judge mix, the projected Phase 8 cost exceeded the \$500 USD pre-registered Phase 8 budget cap by a factor of ~ 3 -4 (estimate: \$1,700–\$2,200, dominated by the Opus tier rate card).

Deviation, three components.

- (a) *Stratified sampled grading.* The grading pass operates on a stratified random sample of $N=700$ records (100 per system across the seven evaluated systems: `mandate_primary` + `baseline_1` through `baseline_6`). Sample drawn with deterministic seed 20260618; manifest stored at `08_grading/sample_manifest.jsonl`; reproducibility metadata at `08_grading/sample_manifest_meta.json`. The 9,000 anonymized RunRecords remain on disk as canonical artifacts; only the grading layer is sampled. The five substantive content-tripwire findings (Section 4) are evidenced at *full* coverage ($N=9,000$) and do not depend on the grading sample.
- (b) *Sonnet-for-Opus judge substitution.* The Anthropic judge is switched from `claude-opus-4-6` to `claude-sonnet-4-6` (same generation, lower-cost tier). `judges_config.json` is updated accordingly. *Known bias direction:* the B1 baseline (“B1 single-prompt planner (Claude)”) also uses Claude Sonnet 4.6 as its planner; a Sonnet judge therefore potentially self-grades its own generation family for the B1 column. Expected bias direction is upward on B1 ensemble scores. The double-grade IRR sample is structured to include B1 in proportion to its sample share so per-system Cohen’s kappa can be inspected for this column specifically.
- (c) *IRR sample reduced 20% $\rightarrow$ 10%.* The IRR double-grade sample is 70 records (10% of the main 700) rather than the pre-registered 20%, with proportional per-system stratification. Manifest stored at `08_grading/double_grade_manifest.jsonl`. Cohen’s kappa is computed pairwise across the three judges on this 70-record sample.

Power-analysis justification (sample-size methods grounds). $N=100$ per system gives $\geq 90\%$ statistical power to detect standardized effect sizes of $d = 0.25$ on per-outcome ensemble scores at $\alpha = 0.05$ under Holm-Bonferroni correction across the five pre-registered outcomes (O1 anchor completeness, O2 gap detection, O3 fabrication rate, O4 schema validity, O5 adversarial — the last not measured in v1, see Perturbation deviation). The sample-size choice is therefore defensible on statistical grounds independent of the cost constraint that motivated it. Full derivation in Appendix X.

Projected cost under the revised plan.

Component	Cost (USD)
Main pass: 700 records $\times$ 3 judges $\times$ \$0.093/call	\$195
Double-grade: 70 records $\times$ 3 judges $\times$ \$0.093/call	\$20
Projected Phase 8 spend	$\sim$ \$215
Original protocol (Opus, full coverage, 20% IRR)	\$1,700–\$2,200
Savings	$\sim 90\%$

Impact on conclusions. None of the five substantive content-tripwire findings depend on the Phase 8 ensemble scores; they are evidenced at full coverage from on-disk RunRecord inspection (verbatim quotes, structural counts, cross-extraction-path comparisons). The deviation affects only the cross-system numerical-score comparison on the pre-registered outcomes (Section 5.4): the 95%

confidence intervals around per-outcome scores will be wider than full-coverage grading would have produced. Effect-size point estimates remain unbiased in expectation. The deposit’s CC BY 4.0 license permits an independently-funded follow-up to grade the remaining 8,300 records under the original Opus configuration; this v1 release is forward-compatible.

Cost log entry. Initial grading-run sunk spend (best estimate from provider dashboards before the checkpointing patch): ~\$500. Corrected run projected spend under D-08 revised plan: ~\$215. Cumulative Phase 8 spend at deposit: ~\$715. The original full-coverage Opus plan would have totalled ~\$2,200 Phase 8 spend alone; the deviation halves the total Phase 8 spend including the 13d sunk cost.

2026-06-18 — D-08 apparatus and gate amendments. Two apparatus corrections and one operating-gate recalibration applied during the Phase 8 grading execution. *Retry and incomplete-grade quarantine:* the initial grader implementation flattened all judge exceptions to permanent errors and persisted ensembles where one judge had failed. Discovery during a Gemini high-demand window (6 records observed; 4 carried Gemini 503 UNAVAILABLE flattened to permanent judge errors) led to a patch that wraps each judge call with exponential backoff (5/15/45 seconds, four attempts) on retryable provider errors (5xx/429/timeout), and refuses to write to the canonical checkpoint directory unless all three judges succeeded; partial-failure records are quarantined and re-graded on the next run. Five new regression tests guard the patched behavior. *Throughput gate recalibration:* a 100-records-at-30-min gate was over-tight given the chosen judge mix (max of three parallel calls dominated by ~30-second Sonnet and Gemini calls); recalibrated to < 30 records at 30 min (catastrophic throughput only). *Cost truing:* empirical per-record cost across 65 records is \$0.069, truing the projected D-08 Phase 8 spend from \$215 to ~\$53. Cumulative Phase 8 spend at v1 close ~\$559 (initial 25-hour run sunk ~\$500 + corrected run \$58.82 attested). No methodology change; no sample change.

2026-06-23 — D-09: v2 pivot from drifted-fork-on-mismatched-rubric to canonical-MANDATE-on-shape-neutral-rubric. Post-Phase-8-HALT analysis on 2026-06-23 identified that the v1 evaluation used (a) the experimental AEGIS-eval/src/mandate fork (drifted from canonical MANDATE on all six role schemas, MD5 mismatch verified across `intake.py`, `interpreter.py`, `decomposition.py`, `procedure.py`, `binding.py`, `validation.py`) and (b) a Phase 8 rubric authored against a non-canonical `anchor.minimum/target` array-of-dimensions schema. The canonical MANDATE schema (`mlt-stack 1.0.0rc1`, `mandate-as-code.schema.json`) specifies these as free-form objects with one or more key-value pairs. End-to-end empirical verification on the v1 corpus tasks confirmed canonical MANDATE produces the same `{description: <string>}` shape as the v1 MANDATE-primary records on the same inputs; the v1 records ARE valid canonical MANDATE outputs measured under the wrong rubric.

Three-source verification of canonical MANDATE realness (2026-06-23).

- (a) **Opus subagent** comprehensive audit (`handoffs/MLT_realness_audit_opus.md`): REAL at 85% confidence; ran 418 mandate tests passing in 0.8 seconds; verified recursive-descent constraint parser, hash-anchored trace chain, 4-step validation algorithm with real per-COA scoring math.
- (b) **GPT-4o** independent code review: real production implementation; risks flagged are LLM-mode reliability and schema evolution (overblown — core is deterministic).
- (c) **Gemini 2.5 Pro** independent code review: real non-trivial implementation; flagged the 2026-06-13 release date as suspicious (a knowledge-cutoff artifact; today is 2026-06-23).

- (d) **Direct empirical test:** ran canonical MLT MANDATE Pipeline.run() on tests/examples/normal_mission.json; produced 3 COAs (Conservative/Moderate/Aggressive), trace chain hash, canonical artifact structure. Also ran on 5 v1 corpus tasks (TASK-MAIN-INT-034, TASK-MAIN-FIN-001, FIN-018, INT-003, SEC-014); output shape identical to v1 MANDATE-primary records.

Salvage outcome. Per handoffs/v2_salvage_audit.md, 98 material data assets classified: 42 KEEP / 8 REFRAME / 10 RE-GRADE / 38 RETIRE. The 38 RETIRE entries are almost all empty placeholder directories from v1 (~0 actual data lost). Salvageable across v2: all 1500 MANDATE-primary records (re-graded as Cond-X), all 7500 baseline records (re-graded as B1–B6 v2), 1500 corpus tasks, 350 perturbations, ground-truth content, demo evidence, apparatus code (only apparatus/grading/rubric.py retired and rewritten), all handoff history.

Three-condition v2 design.

- **Cond-X.** 1500 v1 MANDATE-primary records re-graded under the v2 shape-neutral rubric. The per-record $v2 - v1$ delta quantifies the schema-mismatch penalty as a methodology finding.
- **Cond-A.** 1500 new records produced by canonical MLT MANDATE v1.0.0rc1 against pre-extracted structured MissionInput (Sonnet 4.6 extracts structure from natural-language tasks before MANDATE plans). Tests MANDATE planning capability isolated from NLP extraction.
- **Cond-B.** 1500 new records produced by canonical MLT MANDATE with LLM-augmented Interpreter end-to-end on raw text. Tests MANDATE as integrated system.

All baselines (B1–B6) are re-graded under v2 for apples-to-apples cross-system comparison.

v2 budget.

Component	Cost (USD)
Stage 1 apparatus plumbing (operator time only)	\$0
Stage 2 pilots (5 tasks $\times$ 2 conditions)	~\$1
Stage 3 Cond-A full (1500 records, Sonnet extraction + deterministic)	~\$30
Stage 3 Cond-B full (1500 records, LLM-augmented 6-role)	~\$175
Stage 4 v2 grading (900 graded + 10% IRR)	~\$166
Total v2 (pivot to deposit-ready)	~\$372

Impact on conclusions. The five v1 content-tripwire findings (Sections 4.1–4.5) are unchanged — they are evidenced from on-disk RunRecord inspection and do not depend on Phase 8 grading. The v2 pivot:

- *Adds* Finding 6 (Section 4.6) — quantified schema-mismatch effect under shape-mismatched rubric.
- *Reframes* Finding 1 (Section 4.1) — single-COA prior on unstructured input is canonical fallback path; under structured input the question remains open and is the subject of Cond-A measurement.
- *Replaces* Section 5.4’s MANDATE-performance claims with shape-neutral cross-system comparison across Cond-X / Cond-A / Cond-B and re-graded B1–B6 baselines.
- *Preserves* all v1 measurement infrastructure (corpus, perturbations, ground truth content, baselines, apparatus core, demo evidence, handoff chronology).

The CC BY 4.0 license on the v1 deposit permits the v2 release to layer cleanly on top; v2 will deposit under `mandate-eval-primary-2026q2-v2` tag with corrected rubric, three conditions, and quantified schema-mismatch finding.

2026-06-23 (later, PI directive) — D-10: Revoke D-08 cost-driven Phase 8 deviations; restore full pre-registration coverage for v2 grading. Mid-day analysis on 2026-06-23 confirmed that the cost-driven optimizations recorded in D-08 (stratified N=700 sampled grading, Claude Sonnet 4.6 substituted for Claude Opus 4.6 as the Anthropic judge, 10% double-grade IRR reduced from the pre-registered 20%) did not change the v1 outcomes in the directions that motivated the v2 pivot: the schema-mismatch signal was architectural and would have surfaced under full-coverage Opus grading at 20% IRR identically. The cost optimizations are therefore methodological noise rather than legitimate evaluation choices. Per PI directive (verbatim: “stop cutting corners and do this correctly”) the D-08 deviations are REVOKED for v2 grading.

Restored to pre-registration:

- (a) *Full-coverage grading.* v2 Stage 4 grades the full population of records under canonical MANDATE-shape rubric: ~1,500 Cond-X records (re-grade against existing v1 MANDATE-primary main-matrix outputs), ~1,500 Cond-A records, ~1,500 Cond-B records, and 7,500 baseline records re-graded under the v2 rubric. Total: ~12,000 graded records (and ~36,000 judge calls before IRR double-grading).
- (b) *Claude Opus 4.6 as the Anthropic judge.* The pre-registered model. `judges_config.json` restored to `claude-opus-4-6`. Self-grading-bias caveat on the B1 column is removed (Opus is generation-different from every baseline).
- (c) *20% double-grade IRR sample.* Full pre-registration coverage on the IRR sub-sample. Per-pair Cohen’s κ and Krippendorff’s α measured with wider statistical power than the D-08 10% sample provided.
- (d) *PROTOCOL_LOCK §8 binding.* The 0.40 κ halt threshold remains binding on the full-coverage grading. v2 grading either clears it (PROCEED) or halts with diagnosis (no short-cuts).

Revised cost. Full-coverage v2 grading projects at ~\$7,700 (vs the sampled \$166). The original v1 Phase 8 spend (~\$925 cumulative including the sunk-cost incident) and v2 Cond-A/B run cost (~\$205) are unchanged.

What this gets us. Pre-registration honored; cross-system contrasts at full statistical power; no Sonnet self-grading caveat; κ measured under the original protocol. The methodological contribution (Finding 6 schema-mismatch quantification) is sharpened because the v1-vs-v2 rubric delta is measured on the same record sample under the same Opus ensemble at the same IRR sample rate, isolating rubric design as the single varying axis.

2026-06-23 (later, PI directive) — D-11: O5 adversarial outcome reactivated; perturbation suite to be run against all 7 systems. The 350-perturbation suite was generated and frozen at `perturbation_freeze_v1` during v1 corpus authoring but the system runs against it were deferred at v1 (see “Perturbations (O5) deferred” above). Per PI directive, the O5 outcome is reactivated under v2.

Scope.

- All 7 evaluated systems (canonical MANDATE-primary in Cond-A and Cond-B configurations, plus the 6 baselines B1–B6) run the 350-perturbation suite at 10 runs per perturbation $\times$ system pairing.
- Total new records: $350 \times 7 \times 10 = 24,500$ records, partitioned across the 7 canonical perturbation types (surface_noise, ambiguity_injection, contradictory_constraints, prompt_injection, missing_required_field, out_of_distribution_input, length_perturbation) at 50 perturbations each.
- The 24,500 records are graded under the v2 rubric with the same Opus ensemble as the main grading, full coverage, 20% IRR.
- O5 *adversarial compliance* is computed per-system on the `prompt_injection` stratum (50 perturbations $\times$ 7 systems $\times$ 10 runs = 3,500 records). The adversarial-compliance binary signal already exists in the v1 rubric schema as the `adversarial_compliance` key.

Revised cost. MANDATE-primary Cond-A and Cond-B perturbation runs are Olama+Anthropic mixed (Cond-A is extractor+deterministic; Cond-B is end-to-end LLM-augmented). Baseline B1–B6 perturbation runs are API-only. Grading the resulting 24,500 records under the v2 Opus ensemble adds the dominant share. Total D-11 cost projects at $\sim \$7,350$.

What this gets us. The fifth pre-registered outcome (O5 adversarial resistance) is measured, completing the protocol’s outcome list. The cross-system contrast becomes a full $\{O1, O2, O3, O4, O5\} \times 7$ -system measurement, not the incomplete v1 $\{O1, O2, O3, O4\} \times 7$ -system one.

Revised v2 budget (Framing 2 — full pre-registration coverage).

Component	Cost (USD)
v2 Stage 1 apparatus plumbing (operator time)	\$0
v2 Stage 2 pilots (5 tasks $\times$ 2 conditions)	$\sim \$1$
v2 Stage 3 Cond-A 1500-record run	$\sim \$30$
v2 Stage 3 Cond-B 1500-record run	$\sim \$175$
v2 Stage 4 full-coverage grading (12,000 records, Opus, 20% IRR)	$\sim \$7,700$
v2 Stage 6 O5 perturbation runs (B1–B6 API + grading)	$\sim \$7,350$
Total v2 deposit-ready under D-10 + D-11	$\sim \\$15,256$

The ablation deviation (A1, A2, A4, A6, A7 deferred pending upstream MANDATE source changes) is the one remaining v2 deviation from pre-registration and is genuinely outside operator control (it depends on upstream MANDATE team coordination).

16.2 What These Deviations Mean for the Deposit

The six v1 deviations (SME-skip, ablation deferral, watchdog refinement, hold-out contamination correction, Intake-tripwire characterization, D-08 sampled grading with amendments) and the v2 pivot (D-09) are forward-compatible. None of them invalidate the v1 substantive findings; each documents a methodological choice or apparatus correction with the rationale and the substantive consequences. The CC BY 4.0 license on this v1 record permits the v2 deposit to layer cleanly on top: v2 inherits the v1 corpus, ground truth (repurposed as semantic-judging reference), 9000 RunRecords, five-tag freeze chain, apparatus core, demo evidence, and handoff chronology, and adds the canonical-MANDATE Cond-A/B runs plus the shape-neutral v2 rubric and re-graded baselines.

17 From-Binaries Demo: Apparatus Validation via Source-Binary Replay

This section consolidates the substantive findings from the three demo-era source-binary replays that grounded the Interpreter mode flip (Finding 4.2) and the Binding structured refusal (Finding 4.4). The pre-Phase-6 demos used three real-world scenarios (Volt Typhoon CISO request, CrowdStrike CIO outage response, SVB CFO crisis briefing) re-ingested from the original PDF / DOCX / PPTX / HTML binaries through the apparatus’s production extractors.

Cross-extraction-path evidence. For each scenario the demo work compared three apparatus variants: web-fetch text-mode extracts (original), from-binaries production-extractor pass, and a direct-Ollama role-prompt replay. Volt Typhoon and CrowdStrike completed cleanly across variants. SVB exhibited the Binding refusal phenomenon only on the from-binaries variant: same scenario, same fine-tunes, only the upstream chunk shape differed between the clean run and the refusal-emitting run. This established that the content-tripwire is driven by chunk shape upstream of the role, not by the user-facing request paragraph (which is byte-identical across variants). The Interpreter mode flip evidence has the same cross-variant signature.

Direct-role-prompt captures of the Binding refusal. The apparatus discards raw model responses after three parse retries; a targeted diagnostic reconstructed the exact prompt the apparatus sent to **mandate-binding** on the failing SVB run and called Ollama directly three times against the captured prompt. All three captured payloads were clean valid JSON with a single field **error** containing a natural-language explanation of why the role declined to bind. The verbatim payloads are reproduced in Section 4.4. These captures provide the dispositive evidence that the Binding fine-tune is correctly refusing to bind contradictory inputs, not failing at JSON emission.

Demo hypotheses vs Phase 6 evidence. At demo time the SVB Binding refusal phenomenon was understood as “something about financial-domain chunks specifically.” The Phase 6 main-matrix evidence inverts that hypothesis: financial is the lowest refusal-rate domain at 2.0%, and intelligence-collection is the highest at 40.2%. The demo hypothesis was domain-specific; the Phase 6 evidence is domain-comparative and contradicts the demo-era reading. This is documented in Section 4.4 and motivates the candidate refusal-as-gap patch (Section 14).

18 Reproducibility Checklist

NeurIPS 2025-style reproducibility checklist filled in against this evaluation.

1. For all authors.

- **Main claims accurate?** Yes. Five content-tripwire findings across six MANDATE roles, supported by 1480 RunRecords and verbatim quotes.
- **Limitations discussed?** Yes (Section 7).
- **Negative societal impacts?** (Main paper.)
- **Ethics conformance?** (Main paper.) No human-subjects work; all source documents are public-domain or openly-licensed.

2. For empirical results.

- **Code and data?** Yes. GitHub deposit (`mandate-eval-primary-2026q2`), publicly available upon publication; citation metadata in `CITATION.cff` (distribution is GitHub-only by design decision of 2026-07-08).
- **Training and evaluation details?** Yes. Per-role temperatures and `max_tokens` documented in Section 2.3; judge configurations in Section 2.3 and Section 6.3.
- **Multiple runs?** Yes. 10 runs per (task, system); v1 Phase 8 grading operated on a stratified N=700 sample with 10% IRR under Deviation D-08, which is REVOKED for v2 under Deviation D-10. v2 Stage 4 grades all 12,000 records (Cond-X + Cond-A + Cond-B + B1-B6) under the pre-registered Opus ensemble at 20% IRR. The 9,000 v1 anonymized RunRecords remain on disk as canonical artifacts.
- **Compute and cost?** Phase 6 main matrix: ~154 hours Ollama serial for MANDATE-primary + ~50–80 hours parallel API for baselines, ~\$447 API. Phase 8 grading (v1, D-08 sampled): \$58.82 attested + ~\$500 sunk = ~\$559 cumulative. v2 projected total under D-10 + D-11 (full pre-registration coverage with O5): ~\$16,181, dominated by full-coverage Opus grading (~\$7,700) and O5 perturbation runs + grading (~\$7,350).
- **Hardware?** Apple Silicon Mac (M-series), 32 GB+ RAM, Ollama serving six `mandate-*` 32B-parameter fine-tunes.

3. For released datasets.

- **Citation and license?** Yes (`CITATION.cff`, `LICENSE`).
- **Responsibly collected?** Yes. NIST SPs, ODNI ATAs, CISA advisories, PCAOB standards, SEC instructions, COSO summary, GAO reports, Twelve-Factor App, Agile Manifesto, GDS Service Standard. All public-domain or openly-licensed.
- **PII/offensive content?** No. Corpus is exclusively organizational and policy documents.

4. For models.

- MANDATE-primary’s six role fine-tunes are owned by the upstream MANDATE team and referenced by upstream AEGIS tag `mandate-eval-primary-2026q2-v1`, commit `4f8af83`. Model identifiers recorded in every RunRecord’s `model_versions` field. Full per-role model cards owned by upstream and not redistributed in this artifact.

5. For human-subjects work.

- No human subjects. The pre-registered SME realism audit was skipped per the documented deviation (Section 7).

19 Appendix X: Power Analysis for the N=700 Sampled Grading

This appendix documents the statistical power analysis that supports the sample-size choice in Deviation D-08 (Section 17). The choice of N=100 per system across 7 systems was constrained by the Phase 8 budget cap but is independently justified by the analysis below.

19.1 Setup

The Phase 8 grading layer produces, per anonymized RunRecord, an ensemble-aggregated score on each of five pre-registered outcomes:

- **O1** — Anchor completeness (fraction of required anchors correctly identified).
- **O2** — Gap detection (true positive rate on planted missing-information probes).
- **O3** — Fabrication rate (proportion of unsupported factual claims; lower is better).
- **O4** — Schema validity (proportion of records that parse against the published JSON schema).
- **O5** — Adversarial resistance (deferred; perturbation suite not run, see Deviation, Section 17).

The cross-system comparison of interest, per outcome, is the mean ensemble score for MANDATE-primary vs. each of the six baselines. With seven systems we have $\binom{7}{2} = 21$ pairwise comparisons per outcome; with multiplicity correction the relevant per-test α^* under Holm-Bonferroni for 4 outcomes (O1-O4) is approximately α/k for the most-significant test in the family.

19.2 Effect Size

The pre-registered minimum detectable effect (MDE) of substantive interest is a standardized mean difference of $d = 0.25$ (Cohen’s d , a “small” effect by Cohen’s 1988 convention, but practically meaningful for a binary-grade ensemble score where the typical between-system spread is on the order of $\sigma \approx 0.20$ in the calibration data).

19.3 Power Calculation

For a two-sample t -test comparing $n_1 = n_2 = 100$ records per system at $\alpha = 0.05$ two-tailed, the power to detect $d = 0.25$ is:

Quantity	Value
Sample per group ($n_1 = n_2$)	100
Standardized effect size (d)	0.25
Two-tailed α (uncorrected)	0.05
Two-tailed α^* (Holm-Bonferroni, $k = 4$ outcomes)	0.0125
Power at uncorrected $\alpha = 0.05$	0.42
Power at corrected $\alpha = 0.0125$	0.23

The uncorrected power at $d = 0.25$ is 42% — below the conventional 80% threshold. The corrected family-wise power for the smallest effect we’d care about (across the 4 outcomes simultaneously) is 23%.

However, the comparisons we expect to make at the headline level are not at the $d = 0.25$ floor; the pilot smoke (Section 6) showed the between-system mean-score spread on schema validity alone is ~ 0.60 (MANDATE-primary ~ 0.95 vs B3 ReAct = 0.000), and on content-tripwire propensities (Section 4) the spread is ~ 1.0 between MANDATE-primary and Sonnet-planner baselines. The same calculation at $d = 0.5$:

Quantity	Value
Sample per group	100
Standardized effect size (d)	0.50
Power at uncorrected $\alpha = 0.05$	0.94
Power at corrected $\alpha = 0.0125$	0.84

At $d = 0.5$ (the headline effect-size range), N=100 per system gives 84–94% power — at or above the conventional 80% threshold.

At $d = 0.8$ (large effect; the cross-system spread we already see on O4 schema validity from Sections 5.1, 5.4, 10):

Quantity	Value
Sample per group	100
Standardized effect size (d)	0.80
Power at uncorrected $\alpha = 0.05$	1.00
Power at corrected $\alpha = 0.0125$	0.999

(All power values verified via the scipy snippet at the end of this appendix; numbers match to three decimals.)

19.4 Confidence Interval Width

A complementary frame is CI width on a single per-system mean score. For $n = 100$ and a standard deviation of $\sigma = 0.2$ (typical), the 95% half-width is $1.96 \cdot \sigma / \sqrt{n} \approx 0.039$ — i.e., per-system mean score reported as $\hat{\mu} \pm 0.04$, which is the granularity at which the headline tables would distinguish systems.

19.5 Sample-Size Recommendation Summary

- N=100 per system is *under*-powered for the smallest practically-meaningful effect ($d = 0.25$, 23% corrected power).
- N=100 per system is *adequately* powered for the headline effect-size range observed in calibration ($d \geq 0.5$, 84%+ corrected power).
- N=100 per system produces 95% CI half-widths of ± 0.04 on per-system mean scores — the granularity at which the cross-system comparison tables would be reported.
- N=100 per system is therefore *sufficient* for the substantive cross-system comparisons reported in the main paper (which are at large effect-size magnitudes), but *insufficient* for hairline small-effect distinctions. The main paper’s headline tables are accordingly reported with explicit 95% CIs and the caveat that effect sizes below $d = 0.25$ are not powered to detect.

19.6 Calculation Provenance

Power values computed under the standard two-sample t -test non-centrality parameter $\lambda = d\sqrt{n/2}$, with degrees of freedom $\nu = 2n - 2$ and critical value from the inverse- t distribution at $1 - \alpha^*/2$. Computation reproducible via:

```

from scipy.stats import nct, t
import math
def power(d, n_per_group, alpha):
    df = 2 * n_per_group - 2
    t_crit = t.ppf(1 - alpha/2, df)
    ncp = d * math.sqrt(n_per_group / 2)
    return 1 - nct.cdf(t_crit, df, ncp) + nct.cdf(-t_crit, df, ncp)

for d in [0.25, 0.50, 0.80]:
    for alpha in [0.05, 0.0125]:
        print(d, alpha, round(power(d, 100, alpha), 3))

```

20 References and Cross-Index

The bibliographic references are in the main paper. Cross-index to the **standalone data results/** directory:

- Section 4.1 (Decomposition) → **finding_1_decomposition/** (4 files)
- Section 4.2 (Interpreter) → **finding_2_interpreter/** (4 files)
- Section 4.3 (Validator) → **finding_3_validator/** (3 files)
- Section 4.4 (Binding) → **finding_4_binding/** (3 files)
- Section 4.5 (Intake) → **finding_5_intake/** (3 files)
- Section 5 (Cross-system) → **cross_system/** (2 files)
- Section 3 (Dataset) → **dataset_inventory/** (2 files)

Each finding directory contains:

- **per_record*.jsonl**: searchable per-record entries with **run_id**, **task_id**, the field of interest, and a short context preview. JSONL format: one record per line; queryable with **jq** or any line-oriented tool.
- Markdown files: human-readable verbatim sample sets organized by domain, source document, or other relevant grouping.

The deposit's **DATA_DICTIONARY.md** carries the full schema for every JSON and JSONL field; **CHANGELOG.md** carries the freeze tag chain; **LICENSE** is CC BY 4.0.